FemboyEngine2D API Reference Manual

C++26 data-oriented engine - complete public surface, contracts and a build-from-zero walkthrough

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Contents

- 1. Scope, conformance and engine rules
- 2. Building, targets and compiler flags
- 3. Core value types, FORCEINLINE and the cast family
- 4. FrameArena: the only frame allocator
- 5. JobSystem and per-thread diagnostics
- 6. ECS, scheduling and snapshots
- 7. RHI, render items and frame statistics
- 8. Materials: cooked graphs, the pipeline cache and the shipped path
- 9. Physics: Box2D v3, joints, cloth and SdfCollisionSystem
- 10. NetRollbackSession: deterministic rollback
- 11. Audio: mixer, DSP graph, limiter
- 12. PinkUI and WidgetCanvasSystem
- 13. Accessibility kernel (BlindMode, DeafMode, FlashGuard)
- 14. Asset pipeline, VFS and localization
- 15. Platform services: Steam, Epic Games Store, GOG
- 16. Editor: dashboard, fonts, LogBoy, Scene Diagnostics
- 17. Step-by-step: a game from an empty folder
- 18. Testing, style gate and release checklist
- 19. Open work
- 20. Appendix A - generated symbol index
- 21. Appendix B - macros and platform defines

1. Scope, conformance and engine rules

This manual specifies the public surface of FemboyEngine2D: every header under Source/Engine and the editor infrastructure that a host links against. It is written against the release-candidate tree, so a symbol described here exists in the headers that ship with that tree, and the appendix index is generated from those headers on every run.

The engine targets C++26 with GCC 16 or newer and Clang 19 or newer; the Windows release candidate is built with clang-cl against the MSVC standard library. Four rules are load bearing and every public API is designed around them:

- Simulation state is plain, trivially copyable data. No virtual calls, no owning pointers and no std::string belong in a component, snapshot or network payload.
- Hot paths do not allocate. Frame scratch memory comes from FrameArena, logs go through fixed char buffers, and callbacks are function pointers plus void*, never std::function.
- Determinism is a property of the data, not of the scheduler. Every subsystem exposes a checksummed image, and NetRollbackSession can only replay what it can reproduce byte for byte.
- No RTTI in engine code: dynamic_cast and typeid are rejected by the style gate. Cast<> walks stable type ids instead (see section 3).

The style gate (Source/Tools/Dev/CheckStyle.py) enforces the naming and structure rules: PascalCase file stems, Allman braces, braced control bodies, In-prefixed parameters, Scoped-prefixed locals, no trailing line comments and no Cyrillic in engine sources.

2. Building, targets and compiler flags

One CMake tree produces three kinds of artefact: the engine static library (FemboyEngine2D::Engine), the editor library and application (FemboyEngine2D::Editor), and the offline tools. A shipped game links the engine only; Dear ImGui, ImGuizmo, ImPlot and the TOML writer never enter a shipping target.

CMake option	Default	Effect
FBY_BUILD_EDITOR	ON	Builds fby_editor, the editor app and the panel plugins.
FBY_BUILD_TESTS	OFF	Builds the ctest suite under Source/Tests.
FBY_BUILD_TOOLS	ON	FempakCooker, LocTool, SceneTool, ShaderCompiler, AotCompiler.
FBY_BUILD_PLATFORM_PROVIDERS	ON	Steam and Epic Games Store provider plugins.
FBY_RHI_BACKEND	auto	auto, vulkan, metal or null. auto selects per platform.

```
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=RelWithDebInfo \
      -DFBY_BUILD_TESTS=ON -DFBY_BUILD_EDITOR=ON
```

```
cmake --build build --target fby_editor
ctest --test-dir build --output-on-failure
python Source/Tools/Dev/CheckStyle.py
```

The compiler wrapper (cmake/FbyCompiler.cmake) owns the C++26 flag, the warning set and the sanitizer switches. Do not add flags per target: a target that needs a different dialect is a target that should not exist.

3. Core value types, FORCEINLINE and the cast family

Definitions.hpp is the contract header of the engine. It supplies the export macros, the assertion macros (FBY_ASSERT, FBY_VERIFY, FBY_UNREACHABLE), the stable TypeId machinery, the cast family and the component registration macros. Every other header includes it, directly or transitively.

```
#if defined(_MSC_VER)
#define FORCEINLINE __forceinline
#elif defined(__GNUC__) || defined(__clang__)
#define FORCEINLINE inline __attribute__((always_inline))
#else
#define FORCEINLINE inline
#endif
```

FORCEINLINE is the only inlining spelling in public headers. It is used for accessors, hot arithmetic and template casts; it never replaces a measured optimisation. The three families are:

Symbol	Contract
FemCast<To>(value)	Unchecked static_cast with reference collapsing. Use when the concrete type is proven by construction (factory output, storage tag). No runtime cost.
Cast<To>(object)	Checked cast. Upcast is a compile-time static_cast; downcast compares the object's GetTypeId() against StableTypeId<To>() and its registered parent chain; unrelated types fail to compile. Returns nullptr on a mismatch, never throws, never uses RTTI.
CastComponent<To>(registry, entity)	ECS spelling of Cast<>: EnTT proves the component type at compile time and the registry validates the entity. Returns nullptr when the entity lacks the component.

```
// Downcast through the stable type table, not through RTTI.
Fby::TypeObject* base = factory.Create("Sprite");
if (Fby::Sprite* sprite = Fby::Cast<Fby::Sprite>(base))
{
    sprite->Animate(dt);
}
```

```
// Registration binds Sprite to TypeObject and derives its stable id.
FBY_REGISTER_TYPE(Fby::Sprite, Fby::TypeObject);
```

StableTypeId<T>() is a consteval FNV-1a of the type's registered name (or the compiler signature when no name is pinned). A type that changes its registered name changes its id, which is why component names are part of the serialized schema and why Snapshot v4 stores a LayoutHash next to the name.

4. FrameArena: the only frame allocator

FrameArena is a bump pointer over caller-owned storage. It is reset at the start of a frame and never frees individual allocations. Systems that need transient arrays take it from SystemContext::frameArena; a system that calls operator new in its frame path is a defect, not a style preference.

```
Fby::FrameArena arena(storage);          // caller owns storage
arena.Reset();                           // once per frame
auto* items = arena.AllocateArray<MeshRenderItem>(count);
if (items == nullptr)
{
    // Arena exhausted: degrade, never allocate on the heap.
    return;
}
const std::size_t used = arena.Used();
const std::size_t capacity = arena.Capacity();
```

- Allocate() returns nullptr instead of throwing when the arena is exhausted; callers must handle the failure.
- AllocateArray<T>() static_asserts std::is_trivially_copyable_v<T>: the arena stores POD payloads only.
- Over-aligned types must pass their alignment explicitly; the default is alignof(std::max_align_t).

5. JobSystem and per-thread diagnostics

JobSystem owns a pool of std::jthread workers. Tasks are function pointers plus a context pointer plus a half-open range, so submitting a task never allocates. ParallelFor splits a range into chunks, lets every participant pull the next chunk from an atomic counter, and always runs at least one chunk on the caller, which keeps single-core machines correct and worker re-entry safe.

```
jobs.ParallelFor(entityCount, 4096, [&](std::uint32_t begin, std::uint32_t end)
{
    for (std::uint32_t i = begin; i < end; ++i)
    {
        Integrate(entities[i], dt);
    }
});
jobs.WaitIdle();
```

```
// Diagnostics: one row per thread, slot 0 is the main thread.
std::array<Fby::JobSystem::ThreadLoad, Fby::JobSystem::kMaxThreadSlots> loads{};
const std::uint32_t slots = jobs.SampleThreadLoad(loads);
```

ThreadLoad carries busyNanoseconds (wall time inside engine work), taskCount and a 16-byte name. The counters are one cache line per thread, so two workers finishing at the same time do not contend. SceneDiagnosticsPanel turns two samples into a per-thread occupancy percentage.

ThreadSlotCount() is ThreadCount() + 1 and is clamped by the fixed MCounters array; a device with more than 64 workers reports the first 64 slots.

6. ECS, scheduling and snapshots

The registry is EnTT. Systems are registered on a SystemGraph in one of five phases (PreUpdate, FixedUpdate, Update, PostUpdate, Render). SystemContext is the only parameter a system receives; it carries the registry, JobSystem, frame arena, renderer, window, VFS, AssetDatabase, audio, physics, joints, cloth, fluid, camera stack, accessibility kernel and font atlases.

```
Fby::Ecs::Schedule schedule;
schedule.Add(Fby::Ecs::Phase::FixedUpdate, "physics", &PhysicsStepSystem,
&physics);
schedule.Add(Fby::Ecs::Phase::Update, "movement", &MovementSystem, nullptr);
schedule.Add(Fby::Ecs::Phase::PostUpdate, "platform", &GogAchievementSystem,
&gog);

Fby::Ecs::SystemContext context{registry, jobs};
context.physics = &physics;
context.window = &window;
schedule.Run(Fby::Ecs::Phase::FixedUpdate, context);
```

Snapshot v4 is a canonical, checksummed byte image of the registry. Components are written by hand-written codecs when they contain runtime handles (b2BodyId, b2ShapeId); the snapshot never serializes those handles, and the restore path recreates bodies from the component fields. The schema stores the Mirror name and LayoutHash of every component, so a field change is rejected instead of silently misread.

API	Contract
GetOrAdd<T>(registry, entity)	Adds the component when missing and returns a stable reference.
ecs::SnapshotWriter / SnapshotReader	Writes and validates the canonical image; checksum is FNV-1a over the payload.
Fby::Net::NetRollbackSession	Consumes snapshot v4 and replays ticks on late input (section 10).

7. RHI, render items and frame statistics

The RHI is a hand-written Vulkan/Metal layer. Frames are recorded into persistent ring buffers; sprite items are 48-byte PODs; meshes are drawn through one indirect call per material/layer; linear-blend skinning is one compute dispatch. FrameStats exposes what the pass actually submitted, which is what SceneDiagnosticsPanel plots.

FrameStats field	Meaning
items / batches / drawCalls	Sprite items, batches and submitted draws.
meshItems / meshIndirectCalls	MeshRenderItem count and indirect draws.
skinDispatches / skinRanges	Skinning compute dispatches and ranges.
uploadedBytes	Bytes uploaded this frame.

The editor's Viewport renders into an offscreen target, not into the swapchain. EnsureViewportTarget() sizes that target in physical pixels; a target sized in points is the usual cause of a blurry Viewport on a HiDPI monitor.

8. Materials: cooked graphs, the pipeline cache and the shipped path

Material authoring and material compilation live on opposite sides of the engine boundary. `Render::MaterialGraph` is the authoring DAG: fixed capacities (128 nodes, 384 pins, 64 parameters), trivially copyable, and hashed without the editor positions. `Render::MaterialGraphCompiler` turns a validated graph into GLSL 450 text. `fby_matc` runs that compiler and glslang at build time and writes a `.femmat` payload, so a shipped game never links glslang and never parses GLSL. The full rationale is in [ADR 0002](../../adr/0002-materials-in-engine.md).

Piece	Header	Contract
`Render::MaterialGraph`	`Engine/RHI/MaterialGraph.hpp`	Authoring DAG; `Hash()` is the pipeline identity and excludes UI positions
`MaterialGraphCompiler`	`Engine/RHI/MaterialGraphCompiler.hpp`	DAG to GLSL 450 and the shader contract flags
`EncodeMaterialGraph` / `DecodeMaterialGraph`	`Engine/RHI/MaterialAsset.hpp`	`.fgraph` wire format: explicit little-endian fields, no struct memcpy
`EncodeCookedMaterial` / `DecodeCookedMaterial`	`Engine/RHI/MaterialAsset.hpp`	`.femmat` payload: 32-byte header, vertex and fragment SPIR-V, zero-copy decoding
`Render::MaterialPipelineCache`	`Engine/RHI/MaterialPipelineCache.hpp`	128 pipelines keyed by graph hash, reference counted, LRU eviction, no heap use
`Rhi::RHI::SupportsRuntimeShaderPipelines`	`Engine/RHI/RHI.hpp`	Capability query: Vulkan returns true, Null and Metal return false

```
# Author the graph once, cook it in the build step
build/bin/fby_matc assets/materials/tinted.fgraph -o
assets/materials/tinted.femmat
```

```
// `InBytes` is the .femmat payload from the VFS or a package entry.
Render::CookedMaterialView ScopedMaterial;
if (!Render::DecodeCookedMaterial(InBytes, &ScopedMaterial))
{
    return false;
}

// One cache per device: acquire at load time, release at shutdown.
const Rhi::ShaderPipelineHandle ScopedPipeline = InMaterials.Acquire(InRhi,
ScopedMaterial);

Rhi::MeshRenderItem ScopedItem;
ScopedItem.materialPipeline =
    ScopedPipeline.IsValid() ? ScopedPipeline.id :
Rhi::kDefaultMaterialPipeline;
```

- The graph hash is the pipeline identity. Editing a graph produces a new hash, so the old pipeline becomes an unreferenced cache entry instead of being mutated in place.
- The contract mask travels with the payload: generated vertex stage, fragment view projection, material push constants, world and vertex position varyings. A live entry whose hash arrives with another contract is refused, because that is a different shader.
- `Acquire()` adds a reference and `Release()` drops one. A pipeline at zero references stays resident, so a material acquired once per frame never recreates a pipeline; it is destroyed by LRU eviction, `ReleaseAll()` or `Shutdown()`.
- The cache never allocates and never formats text on the frame path; `LastError()` returns the fixed error buffer.
- Backends without runtime shader pipelines keep the engine shaders. Query `SupportsRuntimeShaderPipelines()` instead of probing for an invalid handle.

Payload	Fields	Validated on load
`.fgraph`	magic `FBYG`, version, node/pin/param counts, nodes, pins, params	magic, version, counts against capacity, exact size, node kinds, pin and param indices
`.femmat`	magic `FBYM`, version, contract mask, graph hash, vertex and fragment byte lengths	magic, version, size against the lengths, both SPIR-V modules, contract consistency

The editor keeps live editing on top of the same cache: `Fby::Editor::MaterialGraphRuntime` recompiles a graph when its hash changes, reports per-tab errors and delegates pipeline ownership to `MaterialPipelineCache`. `MaterialGraphRuntime::SetDomain` switches the live compile between the mesh and the sprite contract, and the panel's `SaveGraph`/`LoadGraph` write and read the

`.fgraph` payload the cooker consumes. A shipped build skips glslang entirely and only calls `DecodeCookedMaterial` and `Acquire`.

Sprites take materials through the third artefact: `MaterialCompileOptions::domain` selects the sprite domain, where the generated fragment stage matches the `quad.vert` varyings (`vUv`, `vColor`, `vLayer`, and `vWorld` when a node needs it) and samples the item's atlas slice through `vLayer`. `Time` is available: the sprite push block carries a material clock at offset 80, so the generated stage reads `pc.materialParams.x` exactly like a mesh shader. `fby_matc --sprite` cooks that variant. `DrawOrder::MaterialBucket` places the sprite in a bucket, `ChangeMaskRenderExtract::SetMaterialPipelines` maps buckets to cached pipeline ids, and the extractor emits one `DrawBatch` per bucket with `DrawBatch::pipeline` already resolved. The 64-bit render key carries the bucket (16 layer, 8 material, 32 depth, 8 sequence), so a material batch is never split by depth. Nodes that need the mesh view-projection (`Fresnel`, `VertexPosition`) are refused in the sprite domain with a readable reason.

9. Physics: Box2D v3, joints, cloth and SdfCollisionSystem

PhysicsSystem owns one b2WorldId and exposes it to extension modules as a packed uint32, so box2d.h never leaks into a public header. Bodies are mirrored from RigidBody2D/Collider2D components into Box2D, stepped at the fixed rate, and read back before interpolation. PhysicsJointSystem builds a fixed-capacity joint graph: chain, rope and spring connections with break limits, damage accumulation, gameplay cuts, repair, in-place rebuild and ties. ClothSystem integrates a Verlet grid and projects it against the same Box2D colliders.

SdfCollisionSystem contract:

Member	Contract
SetField(const SdfField&)	Installs a non-owning mdspan over a row-major grid. Negative values are solid; the field's memory is owned by the renderer/terrain, not by the system.
Step(registry, physics, fixedDt)	Samples the grid at body corners and centre, converts the gradient to a normal, applies a mass-scaled push impulse, reflects inward velocity and writes the corrected state back. Returns the number of corrected bodies.
Update(SystemContext&)	ECS convenience wrapper around Step(); a headless call without physics is a no-op.
Contacts()	Fixed array of SdfContact {entity, point, normal, penetration, impulse, sequence}.
SdfCollisionDesc	pushOut, restitution, maxImpulse, subSamples, dynamicOnly.

```
Fby::SdfCollisionDesc desc;
desc.pushOut = 0.65f;
desc.restitution = 0.05f;
desc.subSamples = 2;
```

```
sdf.Initialize(desc);
sdf.SetField(terrain.Field());
sdf.Register(graph);                    // Phase::FixedUpdate

// Viewport overlay: red vectors whose length is the penetration.
debug.Build(context, Fby::Editor::PhysicsDebugSdf, &sdf);
```

SdfCollisionSystem is not part of SystemContext. A host that wants SDF physics registers the system explicitly through Register(graph) and keeps the instance alive; the Viewport debug pass takes the same pointer.

10. NetRollbackSession: deterministic rollback

A session binds snapshot v4, CommandLog, Pcg32 and the gameplay components (tags, attributes, effects and skills) into one fixed-capacity rollback loop. Everything after Initialize() is allocation-free: one snapshot ring, one input ring per player, a fixed entity map and fixed frame metadata. The simulation callback is a function pointer plus void*.

Method	Contract
Initialize(registry, schema, config, arena)	Allocation-free setup with a caller-owned snapshot arena.
Initialize(registry, schema, config)	Convenience overload; allocates one arena (the only allocation the session makes).
SetSimulation(fn, user) / SetCommandLog(log)	Binds the pure simulation function and the structural command log.
Start(rng)	Captures the base tick and stores the session RNG. Required before Step().
Step(localInput)	Simulates exactly one tick with prediction for missing remote inputs and captures the snapshot.
SubmitRemoteInput(player, frame)	Stores a late or current input; a frame for an already simulated tick marks a pending rollback.
ReplayPending(report)	Restores the snapshot before the earliest corrected tick, restores RNG and CommandLog cursor, re-simulates, captures new snapshots and reports changed ticks.
CompareChecksums(tick, snap, state, out)	Compares a remote report with the stored frame; the two checksums isolate a byte-level desync from a gameplay-state desync.

```
Fby::Net::NetRollbackSession::Config cfg;
cfg.playerCount = 2;
cfg.localPlayer = 0;
cfg.rollbackTicks = 64;
session.Initialize(registry, schema, cfg, arena);
```

```
session.SetSimulation(&SimulateTick, &game);
session.Start(pcg);

while (running)
{
    session.SubmitRemoteInput(remote, received);
    session.Step(localInput);
    if (session.RollbackPending())
    {
        Fby::Net::RollbackReport report;
        session.ReplayPending(&report);
        if (report.changedTicks > 0)
        {
            LogBoy().Log(Fby::Editor::LogBoySeverity::Desync, "net",
                         "rollback {}..{} changed {}", report.fromTick,
report.toTick,
                         report.changedTicks);
        }
    }
}
```

- NetInputFrame is 24 bytes of integer axes (int16 fixed point). Floats are not allowed: a float bit pattern is not guaranteed to survive a different compiler.
- Result enumerates every failure mode (NotInitialized, BufferFull, RollbackTooOld, ChecksumMismatch, ...); ResultName() is the log spelling.
- A session is non-copyable and owns its arena when the convenience overload is used.

11. Audio: mixer, DSP graph, limiter

AudioSystem owns the miniaudio device and a fixed mixer. DspGraph routes nodes into buses, supports ducking and reverb, and receives commands through a lock-free ring so the game thread never blocks the audio callback. Limiter.hpp is the final true-peak stage.

- DeafMode mutes the graph at the Master root and turns sound triggers into subtitle lines; it is a policy of the audio system, not a post-effect.
- Ogg/Opus is decoded off the audio thread; the mixer only sees cooked .fpcm buffers.

12. PinkUI and WidgetCanvasSystem

PinkUI is the in-game UI: a fixed-capacity canvas, mdspan grid layout, anchor resolution, hit testing and one-batch packing. WidgetCanvasSystem turns a WidgetCanvas into FrameOutput (draw items) without allocating; font glyphs come from a .fbyfont atlas mapped through the VFS, so the shipping runtime carries no TTF rasterizer. Glyphs are atlas-only: there is no embedded 8x8 bitmap fallback, and a text frame without a mounted font array is refused instead of drawn with the wrong layer.

Member	Contract
WidgetCanvasSystem::MountFonts(RHI&)	Mandatory cold-start stage on the main thread. Builds the GPU texture array from the

	VFS .fbyfont set; a failure is a PANIC log plus FBY_ASSERT_MSG, never a fallback.
BuildFrame(context, canvas, output)	Lays out and culls one canvas, returning BuildStatus and filling FrameOutput. A text canvas without a current mounted atlas returns FontsNotMounted and writes nothing. Allocation-free; safe to call from Phase::Render.
Update(context, canvases, output)	Builds every active canvas into one output; the host submits the result as one batch. A font-contract violation stops the pass and sets fontContractViolated.
WidgetCanvas::SetDisplayScale	Density from CameraStackSystem::DisplayScale; font sizes are rounded to whole physical pixels.
OledShiftState	Integer 2 px orbit for OLED safety; ApplyOledShift() binds it to a canvas pool.

WidgetCanvasSystem is also the accessibility surface: BlindMode reads widget labels for the narrative log, DeafMode receives subtitle widgets, and colour-vision matrices plus FlashGuard are applied to the canvas colours before packing.

13. Accessibility kernel (BlindMode, DeafMode, FlashGuard)

Accessibility is a frame policy, not an overlay. BlindMode skips the game pass, indirect draws and the skinning dispatch, then publishes a bounded narrative log. DeafMode mutes the Master bus and routes sound triggers into SubtitleLines. FlashGuard compresses sudden luminance changes. Colour-vision correction uses the published Machado 2009 matrices, and SrgbRelativeLuminance()/ContrastRatio() implement WCAG 2.2 for authoring-time validation.

Item	Contract
NarrativeEvent	128-slot SPSC ring of 160-byte text payloads; the simulation thread pushes, the speech thread pops.
ColorVisionMatrix(deficiency)	Returns the 3x3 matrix for protanopia, deuteranopia or tritanopia.
ApplyColorVision(color, matrix, strength)	Applies the matrix in linear light to an sRGB Color32.
FlashGuard::Evaluate(luminance, dt)	Returns the gain for the current frame; a large delta is compressed to the threshold.
EditorSettings::ide*	The same policies applied to the editor window itself: Machado correction, WCAG high-contrast, FlashGuard, keyboard navigation and LogBoy as the narration sink.

14. Asset pipeline, VFS and localization

Assets are addressed by relative path inside the VFS. A .fempak archive is mounted with a priority; a directory mount can shadow it, which is how the editor writes a loose file before the cooker folds it back in. Read() returns a span into an mmap that stays valid until the mount is dropped or the file is hot-reloaded.

Format	Owner	Purpose
.fempak	FempakCooker	Immutable shipped archive with an index and optional encryption.
.fbyfont	FontAtlas (PinkUI)	Baked glyph atlas: [header] [glyph records][pixels]; no runtime TTF rasterizer.
.femeta	AssetDatabase	Typed metadata, import settings and stable AssetId.
.fbin	LocTool	Compiled localization table; the editor UI itself is English-only.

The editor UI is not localizable by design: menus, panel titles, errors and the inspector are English-only. Localization applies to game content and documentation, where a .fbin table and the EN/RU manual cover the shipped languages.

15. Platform services: Steam, Epic Games Store, GOG

PlatformService is the engine-side bridge; PlatformAPI.h is a flat pure-C ABI that a store SDK plugin implements. The provider is loaded dynamically, resolves only documented entry points and converts every store-specific type at the boundary, so a game never includes steam_api.h, EOSSDK or Galaxy headers.

Provider	Status in this tree	Notes
SteamProvider.cpp	Implemented	Steamworks flat API through a function table.
EgsProvider.cpp	Implemented	Epic Online Services provider through the same C ABI.
GogProvider.cpp	Implemented (shim)	Loads galaxy64.dll / libgalaxy64.so / libgalaxy64.dylib at runtime and resolves the flat fby_gog_* shim ABI; the game ships the shim built against its SDK revision. Galaxy is a C++ interface library, so a shim is mandatory. GogAchievementSystem and GogStorageSystem write the

		binary PlatformCommandLog in Phase::PostUpdate.
fby_fontc	Implemented	Offline TTF -> .fbyfont cooker (stb_truetype, RGBA8 atlas, verified header).

```
// Pure C ABI: the provider is a shared library, not a C++ dependency.
typedef struct FbyPlatformApi {
    uint32_t version;
    void* user;
    bool (*init)(void* user);
    void (*shutdown)(void* user);
    void (*unlockAchievement)(void* user, const char* id);
    bool (*writeCloudFile)(void* user, const char* name, const uint8_t* data,
uint32_t size);
} FbyPlatformApi;
```

16. Editor: dashboard, fonts, LogBoy, Scene Diagnostics

The dashboard is the only editor surface that exists before a project is open. It installs the editor font set before the first frame, so the first window is already Inter at the physical DPI. A new project starts in Projects/ under the engine root, resolved by DefaultProjectsRoot() - never in the process working directory and never in the install directory.

Component	Contract
EditorFonts	Loads Inter and JetBrains Mono from the VFS mmap. Subpixel advances (PixelSnapH=false), auto oversampling, no bitmap fallback, dynamic atlas baked at the viewport pixel density. Legacy backends get RasterizerDensity + inverse FontGlobalScale.
ProjectDashboard	Initialize(paths, config, visible, window, vfs, fonts, uiScale): installs typography, resolves the default Projects/ location and creates it.
LogBoySystem / LogBoyPanel	Fixed 4096-record ring of 192-byte text payloads with severity Trace/Info/Warning/Critical/Desync, FNV-1a system hashes and colour+hash filtering.
SceneDiagnosticsPanel	Per-thread occupancy bars and heatmap, FrameArena bytes, MeshRenderItem count, indirect draws, draw calls and exact simulation microseconds with avg/p99/peak.
PhysicsDebugVisualizer	Fixed-buffer line/point geometry for Box2D bodies, ropes, cloth and SDF normals, drawn over the Viewport through a caller-supplied world-to-screen map.

The .fbyfont atlas is a baked PinkUI asset and cannot feed ImGui's vector rasterizer. The editor therefore loads the Inter TTF from the same VFS mount; the atlas header is still validated on startup, so a corrupt game-UI atlas is reported by the editor, not by the player.

17. Step-by-step: a game from an empty folder

This walkthrough produces a playable project with a sprite, physics, keyboard input, a PinkUI HUD and a deterministic rollback session. Every step maps to a real header from the index.

- 1. Launch the editor and create a project with New Project. The dashboard creates <engineRoot>/Projects/<name> with assets/{textures,audio,scenes}, configs/ and project.femproj.
- 2. Import a sprite into assets/textures. The watcher imports it, writes a .femeta and cooks a .ftex texture array layer.
- 3. Create assets/scenes/main.toml with one SpriteRenderer, one Transform2D and one RigidBody2D + Collider2D.
- 4. Register the systems: input in PreUpdate, physics in FixedUpdate, movement in Update, PinkUI in Render.
- 5. Give the camera a CameraStackSystem and call SetDisplayScale() from SDL_GetWindowDisplayScale so UI glyphs are crisp on HiDPI.
- 6. Add an AccessibilityKernel from project settings; BlindMode and DeafMode are read from project.femproj, not hard-coded.
- 7. Create a NetRollbackSession per match, bind a pure SimulateTick callback and feed NetInputFrame values.
- 8. Log through LogBoy: severity + system hash + fixed-format message. Open the LogBoy panel and filter by Desync while testing netcode.
- 9. Enable Physics Debug Draw in the Viewport while tuning colliders and cloth.
- 10. Cook with FempakCooker, run the suite with ctest and the style gate, then ship the engine-only target.

```
// 4-7 in code: systems, accessibility and rollback for one match.
Fby::Ecs::Schedule schedule;
schedule.Add(Fby::Ecs::Phase::PreUpdate, "input", &InputSystem);
schedule.Add(Fby::Ecs::Phase::FixedUpdate, "physics", &PhysicsStepSystem,
&physics);
schedule.Add(Fby::Ecs::Phase::Update, "movement", &MovementSystem);
schedule.Add(Fby::Ecs::Phase::Render, "ui", &UiSystem, &ui);

Fby::AccessibilityKernel kernel;
kernel.Configure(settings);              // blind/deaf/colour/flash

Fby::Net::NetRollbackSession session;
if (session.Initialize(registry, schema, cfg, arena) !=
Fby::Net::NetRollbackSession::Result::Ok)
{
    LogBoy().Log(Fby::Editor::LogBoySeverity::Critical, "net", "rollback init
failed");
```

```
}
```

18. Testing, style gate and release checklist

The suite is deliberately framework-free: every test is an executable with a main() that returns non-zero on failure, so it runs on a headless CI agent, a console devkit and a developer workstation with the same command. The full suite passes at Release.

```
ctest --test-dir build --output-on-failure          # the full suite
python Source/Tools/Dev/CheckStyle.py               # style gate
python Documentation/Tools/GenerateAPIReference.py  # this manual
python Documentation/Tools/GenerateDocumentation.py
python Documentation/Tools/GenerateApplicationGuide.py
```

- A release build enables asserts and runs the whole suite once with FBY_ENABLE_ASSERTS=1, because two latent debug-only compile errors existed in JobSystem at RC.
- The POSIX branch of WriteFileAtomically() and LogLine() must be compiled on Linux/macOS in CI; the Windows-only build never saw them.
- The style gate must report zero failures before a commit is eligible for release.

19. Open work

Open work is tracked in Documentation/ROADMAP.md.

20. Appendix A - generated symbol index

Generated from the headers on every run. Header and line are the authoritative location.

Editor/Infrastructure (115)

Symbol	Kind	Header	Line
AIStateTreeCanvasPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	44
AnimationGraphCanvasPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	45
AnimationTimelinePanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	46
AssetDatabaseWatchSystem	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	20
AssetDatabaseWatchSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	47
AssetImportService	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	30
AssetImportService	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	48
Callbacks	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	38
Change	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	102
Codec	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	220
Command	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	234
CommandKind	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	33

CommandStack	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	30
CommandStack	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	49
Config	Infrastructure	Source/Editor/ Infrastructure/ AssetDatabaseWatch System.hpp	27
Config	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	39
Config	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	88
ContentBrowserPan el	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	50
CookRequest	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	115
CookResult	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	126
EditorCamera	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	91
EditorDesc	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	62
EditorFontRole	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	22
EditorFonts	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	84
EditorFonts	Infrastructure	Source/Editor/ Infrastructure/ EditorTypography.hpp	7
EditorFontSpec	Infrastructure	Source/Editor/ Infrastructure/	38

		EditorFonts.hpp	
EditorFontStatus	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	67
EditorSettings	Infrastructure	Source/Editor/ Infrastructure/ EditorSettings.hpp	19
EditorSettingsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	51
EditorSystem	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	100
Entry	Infrastructure	Source/Editor/ Infrastructure/ TextureThumbnails.hpp	52
ExternalClientDesc	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	37
FrameComposer	Infrastructure	Source/Editor/ Infrastructure/ FrameComposer.hpp	64
FrameComposerStats	Infrastructure	Source/Editor/ Infrastructure/ FrameComposer.hpp	45
FrameGenerationMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	138
FrameGenerationPolicySetting	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	131
HudTargetMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	147
ImFont	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	10
ImGuiIO	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	11
Impl	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	221
InteractionTransacti	Infrastructure	Source/Editor/	16

on		Infrastructure/ EditTransactions.hpp	
LoadedSource	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	138
MaterialGraphPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	52
MaterialGraphRuntime	Infrastructure	Source/Editor/ Infrastructure/ MaterialGraphRuntime.hpp	41
MaterialRuntimeStats	Infrastructure	Source/Editor/ Infrastructure/ MaterialGraphRuntime.hpp	25
MotionHistory2D	Infrastructure	Source/Editor/ Infrastructure/ FrameComposer.hpp	30
NetSession	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	32
NetworkDiagnosticsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	53
OitMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	98
OpaqueRenderMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	91
PendingEdit	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	255
PendingSpawn	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	264
PhysicsDebugFlags	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	19
PhysicsDebugFrame	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	47

PhysicsDebugLine	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	32
PhysicsDebugPoint	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	40
PhysicsDebugVisualizer	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	58
PiePlayerInput	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	27
PlaybackIcon	Infrastructure	Source/Editor/ Infrastructure/ EditorWidgets.hpp	12
PlayInEditor	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	44
Profiler	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	27
ProjectConfig	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	57
ProjectConfig	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	254
ProjectConfigStore	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	54
ProjectConfigStore	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	314
ProjectDashboard	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	55
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	58
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	276

ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	21
ProjectSettings	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	162
ProjectSettings	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	22
ProjectSettingsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	56
Range	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	249
ReleaseCookConfig	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	70
ReleaseCooker	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	154
ReleaseCookProgress	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	117
ReleaseCookResult	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	138
ReleaseCookStage	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	28
ResolvedHudSettings	Infrastructure	Source/Editor/ Infrastructure/ ResolvedRenderSettings.hpp	35
ResolvedRenderSettings	Infrastructure	Source/Editor/ Infrastructure/ ResolvedRenderSettings.hpp	14
Result	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	43
Result	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	47
SdfCollisionSystem	Infrastructure	Source/Editor/	13

		Infrastructure/ PhysicsDebugDraw.hpp	
SdfField	Infrastructure	Source/Editor/ Infrastructure/ FrameComposer.hpp	20
SdfGiVolume	Infrastructure	Source/Editor/ Infrastructure/ FrameComposer.hpp	21
Services	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	220
SettingDesc	Infrastructure	Source/Editor/ Infrastructure/ SettingsSchema.hpp	28
SettingKind	Infrastructure	Source/Editor/ Infrastructure/ SettingsSchema.hpp	20
SettingScope	Infrastructure	Source/Editor/ Infrastructure/ SettingsSchema.hpp	11
Slot	Infrastructure	Source/Editor/ Infrastructure/ MaterialGraphRuntime.hpp	118
Source	Infrastructure	Source/Editor/ Infrastructure/ EditTransactions.hpp	19
SpawnComponent	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	278
SpawnHeader	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	271
SpawnTransaction	Infrastructure	Source/Editor/ Infrastructure/ EditTransactions.hpp	190
StableSource	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	145
Stats	Infrastructure	Source/Editor/ Infrastructure/ AssetDatabaseWatchSystem.hpp	88
Stats	Infrastructure	Source/Editor/	55

		Infrastructure/ AssetImportService.hpp	
Stats	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	95
TemporalUpscaleMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	154
TextureArrayAsset	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hpp	23
TextureBinding	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hpp	142
TextureThumbnailCache	Infrastructure	Source/Editor/ Infrastructure/ TextureThumbnails.hpp	17
UpscaleMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	105
UpscalePolicy	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	114
UpscalerBackendSetting	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	121
UserSettings	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	59
UserSettings	Infrastructure	Source/Editor/ Infrastructure/ UserSettings.hpp	16
VirtualFileSystem	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hpp	20
VirtualFileSystem	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	15
VirtualFileSystem	Infrastructure	Source/Editor/ Infrastructure/	18

		EditorSystem.hpp	
VirtualTextureSettings	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	37
VirtualTextureSettings	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	31
VirtualTextureSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	22
VisualScriptPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	60
WindowManager	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	16

Editor/Panels (79)

Symbol	Kind	Header	Line
Action	Panels	Source/Editor/Panels/ProjectDashboard.hpp	25
AIStateTreeCanvasPanel	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	55
AIStateTreeCompiled	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	35
AIStateTreeDocument	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	24
AnimationGraphCanvasPanel	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	116
AnimationGraphDocument	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	79
AnimationGraphLink	Panels	Source/Editor/Panels/	65

		AnimationGraphCanvasPanel.hpp	
AnimationGraphLinkKind	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	33
AnimationGraphNode	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	45
AnimationGraphNodeKind	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	26
AnimationTimelinePanel	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	34
AssetImportService	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	24
AudioSystem	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	15
AudioSystem	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	11
CanvasModelOps	Panels	Source/Editor/Panels/NodeCanvasPanel.hpp	21
CommandStack	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	25
CommandStack	Panels	Source/Editor/Panels/SettingsPanels.hpp	30
ComponentRegistry	Panels	Source/Editor/Panels/ScriptDebuggerPanel.hpp	18
Config	Panels	Source/Editor/Panels/	35

		ContentBrowserPanel.hpp	
ContentBrowserPanel	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	27
DragPayload	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	60
EditorFonts	Panels	Source/Editor/Panels/ProjectDashboard.hpp	20
EditorFonts	Panels	Source/Editor/Panels/VisualScriptPanel.hpp	20
EditorSettingsPanel	Panels	Source/Editor/Panels/SettingsPanels.hpp	91
EditSource	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	58
EditSource	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	119
Entry	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	43
FileDialog	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	140
ImNodesContext	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	13
ImNodesContext	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	15
LevelSettingsPanel	Panels	Source/Editor/Panels/LevelSettingsPanel.hp	12

		p	
LicensesPanel	Panels	Source/Editor/ Panels/ LicensesPanel.hpp	11
LogBoyPanel	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	394
LogBoyRecord	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	105
LogBoyRing	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	121
LogBoySeverity	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	33
LogBoySystem	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	186
LogicGraphDocument	Panels	Source/Editor/ Panels/ VisualScriptPanel.hpp	23
MaterialGraphPanel	Panels	Source/Editor/ Panels/ MaterialGraphPanel.hpp	20
Mode	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	84
NetSession	Panels	Source/Editor/ Panels/ NetworkDiagnosticsPanel.hpp	6
NetworkDiagnosticsPanel	Panels	Source/Editor/ Panels/ NetworkDiagnosticsPanel.hpp	15
NodeCanvasPanel	Panels	Source/Editor/ Panels/ NodeCanvasPanel.hpp	54
PIEAnimFrame	Panels	Source/Editor/ Panels/ AnimationGraphCanvasPanel.hpp	101

PieClientViewport	Panels	Source/Editor/ Panels/ PieClientViewport.hpp	15
PIEFrameSlot	Panels	Source/Editor/ Panels/ AnimationGraphCanvasPanel.hpp	289
ProjectConfigStore	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	31
ProjectConfigStore	Panels	Source/Editor/ Panels/ VoiceDevicePanel.hpp	17
ProjectDashboard	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	22
ProjectSettingsPanel	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	33
QueryKind	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	430
SceneDiagnosticsPanel	Panels	Source/Editor/ Panels/ SceneDiagnosticsPanel.hpp	56
SceneDiagnosticsSample	Panels	Source/Editor/ Panels/ SceneDiagnosticsPanel.hpp	15
ScriptDebuggerPanel	Panels	Source/Editor/ Panels/ ScriptDebuggerPanel.hpp	28
ScriptRuntime	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	16
SinkUser	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	88
SkeletonInspectorState	Panels	Source/Editor/ Panels/ SkeletonAssetPanel.hpp	19

SpawnContext	Panels	Source/Editor/ Panels/ ContentBrowserPanel. hpp	68
SpriteAnimationInsp ectorState	Panels	Source/Editor/ Panels/ SpriteAnimationPanel. hpp	21
SystemContext	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	24
SystemContext	Panels	Source/Editor/ Panels/ SkeletonAssetPanel.h pp	13
SystemContext	Panels	Source/Editor/ Panels/ SpriteAnimationPanel. hpp	13
SystemName	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	189
TextPreviewPanel	Panels	Source/Editor/ Panels/ TextPreviewPanel.hpp	24
TextureThumbnailCa che	Panels	Source/Editor/ Panels/ SpriteAnimationPanel. hpp	19
TilemapEditorPanel	Panels	Source/Editor/ Panels/ TilemapEditorPanel.h pp	20
TimelinePanelConte xt	Panels	Source/Editor/ Panels/ AnimationTimelinePan el.hpp	22
TransportGroupPane l	Panels	Source/Editor/ Panels/ TransportGroupPanel. hpp	14
UserSettingsPanel	Panels	Source/Editor/ Panels/ UserSettingsPanel.hp p	16
VirtualFileSystem	Panels	Source/Editor/	18

		Panels/ ContentBrowserPanel. hpp	
VirtualFileSystem	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	13
VirtualFileSystem	Panels	Source/Editor/ Panels/ TextPreviewPanel.hpp	15
VirtualTexturePanel	Panels	Source/Editor/ Panels/ VirtualTexturePanel.h pp	18
VirtualTextureSyste m	Panels	Source/Editor/ Panels/ VirtualTexturePanel.h pp	10
VisualScriptPanel	Panels	Source/Editor/ Panels/ ScriptDebuggerPanel. hpp	24
VisualScriptPanel	Panels	Source/Editor/ Panels/ VisualScriptPanel.hpp	32
VoiceDevicePanel	Panels	Source/Editor/ Panels/ VoiceDevicePanel.hpp	19
WidgetCanvas	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	18
WindowManager	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	14

Engine/AI (6)

Symbol	Kind	Header	Line
AIAgentSettings	AI	Source/Engine/AI/ AIController.hpp	16
AIControllerCompon ent	AI	Source/Engine/AI/ AIController.hpp	31
AIControllerSystem	AI	Source/Engine/AI/ AIController.hpp	84

AIPerceptionComponent	AI	Source/Engine/AI/AIController.hpp	59
AIPerceptionSettings	AI	Source/Engine/AI/AIController.hpp	48
AIPerceptionSystem	AI	Source/Engine/AI/AIController.hpp	67

Engine/Assets (58)

Symbol	Kind	Header	Line
AeadAlgorithm	Assets	Source/Engine/Assets/FemCrypto.hpp	18
AeadKey	Assets	Source/Engine/Assets/FemCrypto.hpp	39
AeadNonce	Assets	Source/Engine/Assets/FemCrypto.hpp	46
AeadTag	Assets	Source/Engine/Assets/FemCrypto.hpp	54
ArchiveBuildOptions	Assets	Source/Engine/Assets/FemArchiver.hpp	99
ArchiveCipher	Assets	Source/Engine/Assets/FemArchiver.hpp	29
ArchiveCodec	Assets	Source/Engine/Assets/FemArchiver.hpp	22
ArchiveEntryView	Assets	Source/Engine/Assets/FemArchiver.hpp	115
ArchiveInput	Assets	Source/Engine/Assets/FemArchiver.hpp	72
ArchiveProgress	Assets	Source/Engine/Assets/FemArchiver.hpp	85
ArchiveProgressStage	Assets	Source/Engine/Assets/FemArchiver.hpp	78
ArchiveReader	Assets	Source/Engine/	126

		Assets/ FemArchiver.hpp	
ArchiveReader	Assets	Source/Engine/ Assets/ FemArchiver.hpp	145
ArchiveStatus	Assets	Source/Engine/ Assets/ FemArchiver.hpp	36
AssetDatabase	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	28
AssetDependency	Assets	Source/Engine/ Assets/AssetMeta.hpp	83
AssetHandle	Assets	Source/Engine/ Assets/AssetMeta.hpp	22
AssetImportSettings	Assets	Source/Engine/ Assets/AssetMeta.hpp	45
AssetMeta	Assets	Source/Engine/ Assets/AssetMeta.hpp	61
AssetRecord	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	19
AstcBlock	Assets	Source/Engine/ Assets/ TextureCook.hpp	25
AtlasDesc	Assets	Source/Engine/ Assets/ TextureArray.hpp	31
AtlasLayer	Assets	Source/Engine/ Assets/ TextureArray.hpp	21
AudioClipHeader	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	87
BlockExtent	Assets	Source/Engine/ Assets/ TextureFormat.hpp	103
CleanRenderExtract	Assets	Source/Engine/ Assets/ ProjectScaffold.hpp	127
CookedAudioView	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	116
CookedTexture	Assets	Source/Engine/	41

		Assets/ TextureCook.hpp	
CookedTextureView	Assets	Source/Engine/ Assets/ TextureFormat.hpp	203
CookProgressFlags	Assets	Source/Engine/ Assets/ CookProgress.hpp	31
CookProgressRecord	Assets	Source/Engine/ Assets/ CookProgress.hpp	38
CookProgressStage	Assets	Source/Engine/ Assets/ CookProgress.hpp	18
CookTarget	Assets	Source/Engine/ Assets/ TextureCook.hpp	17
FanmHeader	Assets	Source/Engine/ Assets/ AnimationClipAsset.hpp	22
FemArchiver	Assets	Source/Engine/ Assets/ FemArchiver.hpp	128
FempakInput	Assets	Source/Engine/ Assets/ FempakWriter.hpp	15
FempakWriter	Assets	Source/Engine/ Assets/ FempakWriter.hpp	22
LocaleTable	Assets	Source/Engine/ Assets/ Localization.hpp	14
LocalizationGender	Assets	Source/Engine/ Assets/ Localization.hpp	21
LocalizationPack	Assets	Source/Engine/ Assets/ Localization.hpp	13
LocalizationTable	Assets	Source/Engine/ Assets/ Localization.hpp	28
MappedFile	Assets	Source/Engine/ Assets/VFS.hpp	19
MountRecord	Assets	Source/Engine/	128

		Assets/VFS.hpp	
PakEntry	Assets	Source/Engine/ Assets/ PakFormat.hpp	26
PakHeader	Assets	Source/Engine/ Assets/ PakFormat.hpp	16
PcmFormat	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	21
PinnedFile	Assets	Source/Engine/ Assets/VFS.hpp	63
ReloadNotice	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	86
StagedRow	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	108
StringEntry	Assets	Source/Engine/ Assets/ Localization.hpp	15
TextureAlphaMode	Assets	Source/Engine/ Assets/ TextureFormat.hpp	43
TextureArrayAsset	Assets	Source/Engine/ Assets/ TextureArray.hpp	51
TextureCookOptions	Assets	Source/Engine/ Assets/ TextureCook.hpp	31
TextureFormat	Assets	Source/Engine/ Assets/ TextureFormat.hpp	22
TextureHeader	Assets	Source/Engine/ Assets/ TextureFormat.hpp	184
VfsEntry	Assets	Source/Engine/ Assets/VFS.hpp	55
VirtualFileSystem	Assets	Source/Engine/ Assets/ TextureArray.hpp	16
VirtualFileSystem	Assets	Source/Engine/ Assets/VFS.hpp	69

Engine/Audio (101)

Symbol	Kind	Header	Line
AudioBus	Audio	Source/Engine/Audio/Audio.hpp	26
AudioBusComponent	Audio	Source/Engine/Audio/Audio.hpp	40
AudioCaptureMonitor	Audio	Source/Engine/Audio/AudioDevices.hpp	144
AudioClip	Audio	Source/Engine/Audio/Audio.hpp	69
AudioCommand	Audio	Source/Engine/Audio/DspGraph.hpp	138
AudioCommandType	Audio	Source/Engine/Audio/DspGraph.hpp	121
AudioContextComponent	Audio	Source/Engine/Audio/AudioZones.hpp	67
AudioDeviceEnumerator	Audio	Source/Engine/Audio/AudioDevices.hpp	93
AudioDeviceInfo	Audio	Source/Engine/Audio/AudioDevices.hpp	54
AudioDeviceKind	Audio	Source/Engine/Audio/AudioDevices.hpp	15
AudioDeviceList	Audio	Source/Engine/Audio/AudioDevices.hpp	66
AudioDevicePreferences	Audio	Source/Engine/Audio/AudioDevices.hpp	73
AudioDeviceStatus	Audio	Source/Engine/Audio/AudioDevices.hpp	21
AudioLimiter	Audio	Source/Engine/Audio/Limiter.hpp	23
AudioListener	Audio	Source/Engine/Audio/Audio.hpp	94
AudioSource	Audio	Source/Engine/Audio/Audio.hpp	79
AudioSourceCompo	Audio	Source/Engine/	53

nent		Audio/Audio.hpp	
AudioSystem	Audio	Source/Engine/ Audio/Audio.hpp	100
AudioZoneComponent	Audio	Source/Engine/ Audio/ AudioZones.hpp	32
AudioZoneKind	Audio	Source/Engine/ Audio/ AudioZones.hpp	14
AudioZoneSystem	Audio	Source/Engine/ Audio/ AudioZones.hpp	91
BiquadState	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	369
BusAtomics	Audio	Source/Engine/ Audio/DspGraph.hpp	370
BusFilterParams	Audio	Source/Engine/ Audio/DspGraph.hpp	108
BusRuntime	Audio	Source/Engine/ Audio/Audio.hpp	212
BusShadow	Audio	Source/Engine/ Audio/DspGraph.hpp	291
BusState	Audio	Source/Engine/ Audio/DspGraph.hpp	338
CommandQueue	Audio	Source/Engine/ Audio/DspGraph.hpp	170
CookedClipRecord	Audio	Source/Engine/ Audio/Audio.hpp	187
DspClipView	Audio	Source/Engine/ Audio/DspGraph.hpp	205
DspGraph	Audio	Source/Engine/ Audio/DspGraph.hpp	216
DspGraph	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	16
DspResult	Audio	Source/Engine/ Audio/DspGraph.hpp	34
DspResult	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	19
DuckingParams	Audio	Source/Engine/ Audio/DspGraph.hpp	80

DuckingState	Audio	Source/Engine/ Audio/DspGraph.hpp	298
FilterState	Audio	Source/Engine/ Audio/DspGraph.hpp	327
FloatRing	Audio	Source/Engine/ Audio/Audio.hpp	24
FloatRing	Audio	Source/Engine/ Audio/ AudioStream.hpp	16
FloatRing	Audio	Source/Engine/ Audio/DspGraph.hpp	32
JitterSlot	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	119
LimiterDesc	Audio	Source/Engine/ Audio/Limiter.hpp	11
OggOpusStream	Audio	Source/Engine/ Audio/OggOpus.hpp	131
OggPacket	Audio	Source/Engine/ Audio/OggOpus.hpp	76
OggPacketReader	Audio	Source/Engine/ Audio/OggOpus.hpp	86
OggPage	Audio	Source/Engine/ Audio/OggOpus.hpp	36
OggPageReader	Audio	Source/Engine/ Audio/OggOpus.hpp	57
OpusHeadInfo	Audio	Source/Engine/ Audio/OggOpus.hpp	106
OwnedPcmRecord	Audio	Source/Engine/ Audio/Audio.hpp	194
Peer	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	129
RateBucket	Audio	Source/Engine/ Audio/ VoiceNetworkBridge.hpp	62
ResolvedZone	Audio	Source/Engine/ Audio/ AudioZones.hpp	113
ReverbParams	Audio	Source/Engine/ Audio/DspGraph.hpp	96
ReverbState	Audio	Source/Engine/	311

		Audio/DspGraph.hpp	
SlotRuntime	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	385
SourceRuntime	Audio	Source/Engine/ Audio/Audio.hpp	204
State	Audio	Source/Engine/ Audio/ AudioDevices.hpp	164
Stats	Audio	Source/Engine/ Audio/ AudioZones.hpp	94
VoiceAec	Audio	Source/Engine/ Audio/VoiceAec.hpp	44
VoiceAecSettings	Audio	Source/Engine/ Audio/VoiceAec.hpp	22
VoiceCapture	Audio	Source/Engine/ Audio/ AudioStream.hpp	64
VoiceChannel	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	65
VoiceChannelConfig	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	36
VoiceChannelStats	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	46
VoiceCodecConfig	Audio	Source/Engine/ Audio/ VoiceCodec.hpp	15
VoiceConditionOutput	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	333
VoiceConditionSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	284
VoiceConditionState	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	324
VoiceDecoder	Audio	Source/Engine/ Audio/ VoiceCodec.hpp	51
VoiceDenoiseTier	Audio	Source/Engine/	10

		Audio/VoiceDsp.hpp	
VoiceDistortionSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	157
VoiceDspChain	Audio	Source/Engine/Audio/VoiceDsp.hpp	99
VoiceDspSettings	Audio	Source/Engine/Audio/VoiceDsp.hpp	43
VoiceEffectChain	Audio	Source/Engine/Audio/VoiceEffects.hpp	345
VoiceEffectChainSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	221
VoiceEffectKind	Audio	Source/Engine/Audio/VoiceEffects.hpp	20
VoiceEffectPresetKind	Audio	Source/Engine/Audio/VoiceEffects.hpp	235
VoiceEffectProfile	Audio	Source/Engine/Audio/VoiceEffects.hpp	310
VoiceEffectRack	Audio	Source/Engine/Audio/VoiceEffects.hpp	433
VoiceEffectSlot	Audio	Source/Engine/Audio/VoiceEffects.hpp	200
VoiceEncoder	Audio	Source/Engine/Audio/VoiceCodec.hpp	27
VoiceFormantSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	146
VoiceMuffleSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	80
VoiceMuteSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	67
VoiceNetworkBridge	Audio	Source/Engine/Audio/VoiceNetworkBridge.h	18

		pp	
VoiceParticipant	Audio	Source/Engine/Audio/VoiceSession.hpp	46
VoicePitchSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	133
VoicePresetKind	Audio	Source/Engine/Audio/VoiceDsp.hpp	68
VoiceRadioSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	115
VoiceRingModulationSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	171
VoiceRouting	Audio	Source/Engine/Audio/VoiceChannel.hpp	30
VoiceSession	Audio	Source/Engine/Audio/VoiceChannel.hpp	15
VoiceSession	Audio	Source/Engine/Audio/VoiceSession.hpp	65
VoiceSpatialPosition	Audio	Source/Engine/Audio/VoiceSession.hpp	37
VoiceSpatialSettings	Audio	Source/Engine/Audio/VoiceSession.hpp	22
VoiceState	Audio	Source/Engine/Audio/DspGraph.hpp	350
VoiceSystem	Audio	Source/Engine/Audio/VoiceSystem.hpp	39
VoiceSystemConfig	Audio	Source/Engine/Audio/VoiceSystem.hpp	22
VoiceUnderwaterSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	98
VoiceWhisperSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	185

VorbisIdentInfo	Audio	Source/Engine/Audio/OggOpus.hpp	121

Engine/Core (103)

Symbol	Kind	Header	Line
AssetId	Core	Source/Engine/Core/AssetId.hpp	43
AssetKind	Core	Source/Engine/Core/AssetId.hpp	15
BigIntT	Core	Source/Engine/Core/BigInt.hpp	179
BigNumberStore	Core	Source/Engine/Core/BigNumberStore.hpp	41
BinaryKind	Core	Source/Engine/Core/BigNumberStore.hpp	259
BinaryKind	Core	Source/Engine/Core/DecimalStore.hpp	206
BufferTag	Core	Source/Engine/Core/Types.hpp	301
BuildInfo	Core	Source/Engine/Core/BuildInfo.hpp	12
BuildProfile	Core	Source/Engine/Core/BuildProfile.hpp	135
CameraView	Core	Source/Engine/Core/Grid.hpp	337
Cell	Core	Source/Engine/Core/Grid.hpp	74
CivilDate	Core	Source/Engine/Core/Calendar.hpp	21
Clock	Core	Source/Engine/Core/Time.hpp	28
ClockTime	Core	Source/Engine/Core/Calendar.hpp	32
Color32	Core	Source/Engine/Core/Types.hpp	203
ComponentDesc	Core	Source/Engine/Core/Mirror.hpp	48
ComponentRegistry	Core	Source/Engine/Core/PluginHost.hpp	27
ComponentType	Core	Source/Engine/Core/	18

		PluginHost.hpp	
DateTime	Core	Source/Engine/Core/Calendar.hpp	44
DecimalStore	Core	Source/Engine/Core/DecimalStore.hpp	17
DecimalT	Core	Source/Engine/Core/Decimal.hpp	34
DurationStyle	Core	Source/Engine/Core/Time.hpp	571
EddsaKeyPair	Core	Source/Engine/Core/CryptoAsymmetric.hpp	31
FieldDesc	Core	Source/Engine/Core/Mirror.hpp	37
FieldKind	Core	Source/Engine/Core/Mirror.hpp	17
FileFilter	Core	Source/Engine/Core/WindowManager.hpp	183
FormatSpec	Core	Source/Engine/Core/BigInt.hpp	36
FrameArena	Core	Source/Engine/Core/FrameArena.hpp	12
FrameTimer	Core	Source/Engine/Core/Time.hpp	501
GamepadState	Core	Source/Engine/Core/WindowManager.hpp	26
GridDesc	Core	Source/Engine/Core/Grid.hpp	91
Handle	Core	Source/Engine/Core/BigNumberStore.hpp	20
Handle	Core	Source/Engine/Core/Types.hpp	275
Handle	Core	Source/Engine/Core/WindowHost.hpp	20
Host	Core	Source/Engine/Core/Plugin.hpp	20
InputState	Core	Source/Engine/Core/WindowManager.hpp	114
JobSystem	Core	Source/Engine/Core/	25

		JobSystem.hpp	
Key	Core	Source/Engine/ Core/ WindowManager.hpp	46
KeyLess	Core	Source/Engine/Core/ FlatMap.hpp	203
LoadedPlugin	Core	Source/Engine/Core/ PluginHost.hpp	64
LogLevel	Core	Source/Engine/Core/ Log.hpp	13
LogRetention	Core	Source/Engine/Core/ Log.hpp	31
Mat4	Core	Source/Engine/ Core/Mat4.hpp	14
NoiseGenerator	Core	Source/Engine/Core/ NoiseGenerator.hpp	10
Notation	Core	Source/Engine/Core/ BigInt.hpp	30
NoticeEntry	Core	Source/Engine/Core/ ThirdPartyNotices.hpp	12
OptimizationAdvice	Core	Source/Engine/Core/ BuildProfile.hpp	154
OptimizationLevel	Core	Source/Engine/Core/ BuildProfile.hpp	10
Pcg32	Core	Source/Engine/Core/ Pcg32.hpp	20
PipelineTag	Core	Source/Engine/Core/ Types.hpp	303
PluginHost	Core	Source/Engine/Core/ PluginHost.hpp	76
PluginSystem	Core	Source/Engine/Core/ PluginHost.hpp	54
Profiler	Core	Source/Engine/Core/ Profiler.hpp	43
ProfileRow	Core	Source/Engine/Core/ ProfilerPanel.hpp	26
ProfilerPanel	Core	Source/Engine/Core/ ProfilerPanel.hpp	39
ProfileSample	Core	Source/Engine/Core/ Profiler.hpp	28
ProfileSortMode	Core	Source/Engine/Core/	16

		ProfilerPanel.hpp	
ProfileStats	Core	Source/Engine/Core/ Profiler.hpp	16
ProjectionMode	Core	Source/Engine/Core/ Grid.hpp	17
Rect	Core	Source/Engine/Core/ Types.hpp	173
Reflection	Core	Source/Engine/ Core/Mirror.hpp	66
Reflection	Core	Source/Engine/ Core/Mirror.hpp	294
Reflection	Core	Source/Engine/ Core/Mirror.hpp	306
ReplayFrame	Core	Source/Engine/Core/ Replay.hpp	238
ReplayKeyframe	Core	Source/Engine/Core/ Replay.hpp	16
ReplaySession	Core	Source/Engine/Core/ Replay.hpp	25
RngState	Core	Source/Engine/Core/ Pcg32.hpp	11
RoundingMode	Core	Source/Engine/Core/ Decimal.hpp	22
SamplerTag	Core	Source/Engine/Core/ Types.hpp	304
Scope	Core	Source/Engine/Core/ Profiler.hpp	141
SecureRng	Core	Source/Engine/Core/ SecureRng.hpp	11
Sha256Context	Core	Source/Engine/Core/ Sha256.hpp	17
Slot	Core	Source/Engine/Core/ BigNumberStore.hpp	252
Slot	Core	Source/Engine/Core/ DecimalStore.hpp	199
SplashClock	Core	Source/Engine/Core/ Splash.hpp	10
Status	Core	Source/Engine/Core/ BigInt.hpp	18
Stopwatch	Core	Source/Engine/Core/ Time.hpp	209

Task	Core	Source/Engine/Core/JobSystem.hpp	96
TextureTag	Core	Source/Engine/Core/Types.hpp	302
TextWriter	Core	Source/Engine/Core/Calendar.hpp	189
ThreadCounter	Core	Source/Engine/Core/JobSystem.hpp	109
ThreadLoad	Core	Source/Engine/Core/JobSystem.hpp	31
TimerFired	Core	Source/Engine/Core/Time.hpp	255
TimerHandle	Core	Source/Engine/Core/Time.hpp	240
TimerPool	Core	Source/Engine/Core/Time.hpp	266
TypeNameOverride	Core	Source/Engine/Core/Definitions.hpp	40
TypeNameOverride	Core	Source/Engine/Core/Definitions.hpp	283
TypeObject	Core	Source/Engine/Core/Definitions.hpp	185
TypeRecord	Core	Source/Engine/Core/Definitions.hpp	85
TypeTable	Core	Source/Engine/Core/Definitions.hpp	93
TypeTraits	Core	Source/Engine/Core/Definitions.hpp	170
TypeTraits	Core	Source/Engine/Core/Definitions.hpp	267
Unregistered	Core	Source/Engine/Core/Mirror.hpp	59
Vector2D	Core	Source/Engine/Core/Types.hpp	13
Vector3D	Core	Source/Engine/Core/Types.hpp	90
Vector4D	Core	Source/Engine/Core/Types.hpp	139
WindowDesc	Core	Source/Engine/Core/WindowManager.hpp	204
WindowExtent	Core	Source/Engine/	37

		Core/ WindowManager.hpp	
WindowHost	Core	Source/Engine/ Core/WindowHost.hpp	17
WindowManager	Core	Source/Engine/ Core/ WindowManager.hpp	215
WindowMode	Core	Source/Engine/ Core/ WindowManager.hpp	195
WindowRecord	Core	Source/Engine/ Core/WindowHost.hpp	30
X25519KeyPair	Core	Source/Engine/Core/ CryptoAsymmetric.hp p	20

Engine/ECS (211)

Symbol	Kind	Header	Line
AccessibilityKernel	ECS	Source/Engine/ECS/ Schedule.hpp	21
AICommandCompon ent	ECS	Source/Engine/ECS/ StateTree.hpp	175
AICommandKind	ECS	Source/Engine/ECS/ StateTree.hpp	163
AnimationTimeline	ECS	Source/Engine/ECS/ AnimationTimeline.hp p	90
AnimBoneKey	ECS	Source/Engine/ECS/ AnimGraph.hpp	74
AnimClip	ECS	Source/Engine/ECS/ AnimGraph.hpp	86
AnimConditionFlags	ECS	Source/Engine/ECS/ AnimGraph.hpp	28
AnimGraphSystem	ECS	Source/Engine/ECS/ AnimGraph.hpp	180
AnimRestBone	ECS	Source/Engine/ECS/ AnimGraph.hpp	104
AnimStateCompone nt	ECS	Source/Engine/ECS/ AnimGraph.hpp	114
AnimTransition	ECS	Source/Engine/ECS/ AnimGraph.hpp	48

AssetDatabase	ECS	Source/Engine/ECS/Scene.hpp	21
AssetDatabase	ECS	Source/Engine/ECS/Schedule.hpp	33
AudioSystem	ECS	Source/Engine/ECS/Schedule.hpp	16
BigIntComponent	ECS	Source/Engine/ECS/Components.hpp	265
CachedTarget	ECS	Source/Engine/ECS/LightSource.hpp	229
Camera2D	ECS	Source/Engine/ECS/Components.hpp	218
CameraStackSystem	ECS	Source/Engine/ECS/Schedule.hpp	26
CarrierMemberComponent	ECS	Source/Engine/ECS/Components.hpp	168
ChainIdTag	ECS	Source/Engine/ECS/StableIds.hpp	12
ChainSaveV3	ECS	Source/Engine/ECS/SnapshotWorld.hpp	28
ChangeMaskBit	ECS	Source/Engine/ECS/ChangeMask.hpp	19
CharacterDesc	ECS	Source/Engine/ECS/Descriptors.hpp	51
ChunkCopyRange	ECS	Source/Engine/ECS/RenderEcs.hpp	90
ChunkHeader	ECS	Source/Engine/ECS/RenderEcs.hpp	45
ClothIdTag	ECS	Source/Engine/ECS/StableIds.hpp	13
ClothSaveV3	ECS	Source/Engine/ECS/SnapshotWorld.hpp	59
ClothSystem	ECS	Source/Engine/ECS/Schedule.hpp	23
ColliderSnapshot	ECS	Source/Engine/ECS/Snapshot.hpp	181
CollisionMode	ECS	Source/Engine/ECS/Presentation.hpp	21
CollisionPolicyComponent	ECS	Source/Engine/ECS/Presentation.hpp	101
Command	ECS	Source/Engine/ECS/	29

		CommandBuffer.hpp	
CommandBuffer	ECS	Source/Engine/ECS/ CommandBuffer.hpp	37
CommandBuffer	ECS	Source/Engine/ECS/ Schedule.hpp	54
CommandCutChain	ECS	Source/Engine/ECS/ CommandLog.hpp	42
CommandKind	ECS	Source/Engine/ECS/ CommandBuffer.hpp	19
CommandLog	ECS	Source/Engine/ECS/ CommandLog.hpp	92
CommandLogHeader	ECS	Source/Engine/ECS/ CommandLog.hpp	66
CommandRecord	ECS	Source/Engine/ECS/ CommandLog.hpp	78
CommandType	ECS	Source/Engine/ECS/ CommandLog.hpp	22
CommandWorldCallbacks	ECS	Source/Engine/ECS/ CommandLog.hpp	52
ComponentChangeMask	ECS	Source/Engine/ECS/ ChangeMask.hpp	52
ComponentOps	ECS	Source/Engine/ECS/ ComponentRegistry.hpp	30
ComponentOpsFor	ECS	Source/Engine/ECS/ ComponentRegistry.hpp	55
ComponentRegistry	ECS	Source/Engine/ECS/ ComponentRegistry.hpp	116
CompressedClip	ECS	Source/Engine/ECS/ AnimationCompression.hpp	47
CompressedKey	ECS	Source/Engine/ECS/ AnimationCompression.hpp	19
CompressedTrack	ECS	Source/Engine/ECS/ AnimationCompression.hpp	31
CompressionReport	ECS	Source/Engine/ECS/ AnimationCompression.hpp	63
Cursor	ECS	Source/Engine/ECS/	177

		CommandLog.hpp	
DecimalComponent	ECS	Source/Engine/ECS/Components.hpp	276
DrawOrder	ECS	Source/Engine/ECS/Components.hpp	207
EditorSelected	ECS	Source/Engine/ECS/Components.hpp	232
EmitterIdTag	ECS	Source/Engine/ECS/StableIds.hpp	14
EmitterSaveV3	ECS	Source/Engine/ECS/SnapshotWorld.hpp	79
EntityRemap	ECS	Source/Engine/ECS/Scene.hpp	59
Entry	ECS	Source/Engine/ECS/Schedule.hpp	130
FeskIHeader	ECS	Source/Engine/ECS/SkeletonAsset.hpp	37
FeskINameRef	ECS	Source/Engine/ECS/SkeletonAsset.hpp	87
FeskISprite	ECS	Source/Engine/ECS/SkeletonAsset.hpp	97
FluidSaveV3	ECS	Source/Engine/ECS/SnapshotWorld.hpp	71
FluidWaveSystem	ECS	Source/Engine/ECS/Schedule.hpp	24
FontAtlasSet	ECS	Source/Engine/ECS/Schedule.hpp	41
FrameArena	ECS	Source/Engine/ECS/Schedule.hpp	17
GameplayAttributeRegistry	ECS	Source/Engine/ECS/Schedule.hpp	28
GameplayEffectSystem	ECS	Source/Engine/ECS/Schedule.hpp	29
GameplaySkillSystem	ECS	Source/Engine/ECS/Schedule.hpp	30
GameplayTagRegistry	ECS	Source/Engine/ECS/Schedule.hpp	27
GpuEntity	ECS	Source/Engine/ECS/RenderEcs.hpp	31
GpuSpriteInstance	ECS	Source/Engine/ECS/RenderEcs.hpp	245

GraphSystem	ECS	Source/Engine/ECS/SystemGraph.hpp	17
Hidden	ECS	Source/Engine/ECS/Components.hpp	237
IkChain	ECS	Source/Engine/ECS/SkeletonIk.hpp	43
IkComponent	ECS	Source/Engine/ECS/SkeletonIk.hpp	64
IkGoalComponent	ECS	Source/Engine/ECS/SkeletonIk.hpp	77
IkSolver	ECS	Source/Engine/ECS/SkeletonIk.hpp	19
InputUserComponent	ECS	Source/Engine/ECS/Components.hpp	65
Interpolation	ECS	Source/Engine/ECS/AnimationTimeline.hpp	29
ITextShaper	ECS	Source/Engine/ECS/Schedule.hpp	47
Lifetime	ECS	Source/Engine/ECS/Components.hpp	226
Light2DComponent	ECS	Source/Engine/ECS/Lighting.hpp	37
Light2DOccluderComponent	ECS	Source/Engine/ECS/Lighting.hpp	106
Light2DResultComponent	ECS	Source/Engine/ECS/Lighting.hpp	69
Light2DTargetComponent	ECS	Source/Engine/ECS/Lighting.hpp	53
LightGrid2D	ECS	Source/Engine/ECS/LightGrid.hpp	26
LightGridInput	ECS	Source/Engine/ECS/LightGrid.hpp	16
LightingSettings	ECS	Source/Engine/ECS/Lighting.hpp	117
LightingSystem	ECS	Source/Engine/ECS/Lighting.hpp	144
LightSourceComponent	ECS	Source/Engine/ECS/LightSource.hpp	48
LightSourceFlags	ECS	Source/Engine/ECS/LightSource.hpp	32
LightSourceSettings	ECS	Source/Engine/ECS/	149

		LightSource.hpp	
LightSourceSystem	ECS	Source/Engine/ECS/ LightSource.hpp	168
LightSourceType	ECS	Source/Engine/ECS/ LightSource.hpp	22
LocalizationTable	ECS	Source/Engine/ECS/ Schedule.hpp	22
LocalizedTextCompo nent	ECS	Source/Engine/ECS/ Components.hpp	323
LocalTransform2D	ECS	Source/Engine/ECS/ Components.hpp	34
Mat3	ECS	Source/Engine/ECS/ Skeleton.hpp	31
NameComponent	ECS	Source/Engine/ECS/ Components.hpp	244
NarrativeLabelComp onent	ECS	Source/Engine/ECS/ Components.hpp	254
NativeCleanupHook	ECS	Source/Engine/ECS/ RegistryOps.hpp	12
PackedTransform2D	ECS	Source/Engine/ECS/ RenderEcs.hpp	229
Parent2D	ECS	Source/Engine/ECS/ Components.hpp	42
Phase	ECS	Source/Engine/ECS/ Schedule.hpp	59
PhysicsBodyDesc	ECS	Source/Engine/ECS/ Descriptors.hpp	22
PhysicsJointSystem	ECS	Source/Engine/ECS/ Schedule.hpp	25
PhysicsShapeKind	ECS	Source/Engine/ECS/ Descriptors.hpp	15
PhysicsSystem	ECS	Source/Engine/ECS/ Schedule.hpp	18
PrefabAsset	ECS	Source/Engine/ECS/ Descriptors.hpp	88
PrefabEntityDesc	ECS	Source/Engine/ECS/ Descriptors.hpp	76
PrefabEntityFlags	ECS	Source/Engine/ECS/ Descriptors.hpp	98
PrefabEntityOverride	ECS	Source/Engine/ECS/ PrefabSystem.hpp	18

PrefabOverrides	ECS	Source/Engine/ECS/PrefabSystem.hpp	31
PrefabSystem	ECS	Source/Engine/ECS/PrefabSystem.hpp	36
QuantizedPose	ECS	Source/Engine/ECS/SkeletonIk.hpp	165
RenderChunk	ECS	Source/Engine/ECS/RenderEcs.hpp	62
RenderChunkArena	ECS	Source/Engine/ECS/RenderEcs.hpp	304
RenderEcs	ECS	Source/Engine/ECS/RenderEcs.hpp	456
RenderMaskComponent	ECS	Source/Engine/ECS/Presentation.hpp	139
RenderPassMask	ECS	Source/Engine/ECS/Presentation.hpp	126
ReplayPlayback	ECS	Source/Engine/ECS/ReplayRecorder.hpp	103
ReplayRecorder	ECS	Source/Engine/ECS/ReplayRecorder.hpp	32
RetargetMap	ECS	Source/Engine/ECS/SkeletonIk.hpp	131
RetargetPair	ECS	Source/Engine/ECS/SkeletonIk.hpp	118
RHI	ECS	Source/Engine/ECS/Schedule.hpp	37
RigidBodySnapshot	ECS	Source/Engine/ECS/Snapshot.hpp	167
SceneComponent	ECS	Source/Engine/ECS/Scene.hpp	38
SceneData	ECS	Source/Engine/ECS/Scene.hpp	52
SceneEntity	ECS	Source/Engine/ECS/Scene.hpp	45
SceneLoadOptions	ECS	Source/Engine/ECS/Scene.hpp	76
SceneMeta	ECS	Source/Engine/ECS/Scene.hpp	31
Schedule	ECS	Source/Engine/ECS/Schedule.hpp	125
SdfBrickMap	ECS	Source/Engine/ECS/	17

		Lighting.hpp	
SdfGiVolume	ECS	Source/Engine/ECS/Lighting.hpp	18
SdfLightingFrame	ECS	Source/Engine/ECS/Lighting.hpp	22
SdfSunComponent	ECS	Source/Engine/ECS/Lighting.hpp	81
SkeletonAssetRef	ECS	Source/Engine/ECS/SkeletonAssetRef.hpp	20
SkeletonAssetSource	ECS	Source/Engine/ECS/SkeletonAsset.hpp	189
SkeletonAssetView	ECS	Source/Engine/ECS/SkeletonAsset.hpp	121
SkeletonBoneName	ECS	Source/Engine/ECS/SkeletonAsset.hpp	168
SkeletonComponent	ECS	Source/Engine/ECS/Skeleton.hpp	121
SkeletonNameTable	ECS	Source/Engine/ECS/SkeletonAsset.hpp	175
SkeletonSystem	ECS	Source/Engine/ECS/Skeleton.hpp	198
SkinComponent	ECS	Source/Engine/ECS/Skeleton.hpp	136
SkinnedSprite	ECS	Source/Engine/ECS/Skeleton.hpp	147
SleepingStaticComponent	ECS	Source/Engine/ECS/Presentation.hpp	202
SnapshotBaseline	ECS	Source/Engine/ECS/Snapshot.hpp	314
SnapshotComponent	ECS	Source/Engine/ECS/Snapshot.hpp	76
SnapshotComponentOp	ECS	Source/Engine/ECS/Snapshot.hpp	41
SnapshotExtras	ECS	Source/Engine/ECS/Snapshot.hpp	265
SnapshotFlags	ECS	Source/Engine/ECS/Snapshot.hpp	29
SnapshotHeader	ECS	Source/Engine/ECS/Snapshot.hpp	47
SnapshotRecordFlags	ECS	Source/Engine/ECS/Snapshot.hpp	35

SnapshotRestoreExtras	ECS	Source/Engine/ECS/Snapshot.hpp	270
SnapshotSchema	ECS	Source/Engine/ECS/Snapshot.hpp	91
SnapshotSchema	ECS	Source/Engine/ECS/Snapshot.hpp	206
SnapshotView	ECS	Source/Engine/ECS/Snapshot.hpp	255
Sprite2D	ECS	Source/Engine/ECS/Components.hpp	192
SpriteAnimationComponent	ECS	Source/Engine/ECS/SpriteAnimation.hpp	21
SpriteAnimationEventQueue	ECS	Source/Engine/ECS/SpriteAnimation.hpp	74
SpriteAnimationSystem	ECS	Source/Engine/ECS/SpriteAnimation.hpp	118
SpriteFrameEvent	ECS	Source/Engine/ECS/SpriteAnimation.hpp	66
SpriteFrameRect	ECS	Source/Engine/ECS/SpriteAnimation.hpp	55
SpriteSheetComponent	ECS	Source/Engine/ECS/SpriteAnimation.hpp	40
StableIdAllocator	ECS	Source/Engine/ECS/StableIds.hpp	25
StateTreeComponent	ECS	Source/Engine/ECS/StateTree.hpp	192
StateTreeDefinition	ECS	Source/Engine/ECS/StateTree.hpp	152
StateTreeEvaluator	ECS	Source/Engine/ECS/StateTree.hpp	55
StateTreeEvaluatorKind	ECS	Source/Engine/ECS/StateTree.hpp	28
StateTreeNode	ECS	Source/Engine/ECS/StateTree.hpp	115
StateTreeNodeKind	ECS	Source/Engine/ECS/StateTree.hpp	105
StateTreeSystem	ECS	Source/Engine/ECS/StateTree.hpp	237
StateTreeTask	ECS	Source/Engine/ECS/StateTree.hpp	87
StateTreeTaskKind	ECS	Source/Engine/ECS/	73

		StateTree.hpp	
StateTreeTransition	ECS	Source/Engine/ECS/StateTree.hpp	134
SystemContext	ECS	Source/Engine/ECS/Schedule.hpp	87
SystemGraph	ECS	Source/Engine/ECS/Schedule.hpp	55
SystemGraph	ECS	Source/Engine/ECS/SystemGraph.hpp	29
TextComponent	ECS	Source/Engine/ECS/Components.hpp	295
TextConditionComponent	ECS	Source/Engine/ECS/Components.hpp	379
TextEffectComponent	ECS	Source/Engine/ECS/Components.hpp	349
TextFormatComponent	ECS	Source/Engine/ECS/Components.hpp	361
TextService	ECS	Source/Engine/ECS/Schedule.hpp	45
TextShapeCache	ECS	Source/Engine/ECS/Schedule.hpp	46
TextShapeComponent	ECS	Source/Engine/ECS/Components.hpp	338
TilemapAssetRefComponent	ECS	Source/Engine/ECS/Tilemap.hpp	130
TilemapComponent	ECS	Source/Engine/ECS/Tilemap.hpp	63
TilemapRenderSystem	ECS	Source/Engine/ECS/Tilemap.hpp	150
TileQuad	ECS	Source/Engine/ECS/Tilemap.hpp	142
TimelineEvent	ECS	Source/Engine/ECS/AnimationTimeline.hpp	67
TimelineEventKind	ECS	Source/Engine/ECS/AnimationTimeline.hpp	36
TimelineKey	ECS	Source/Engine/ECS/AnimationTimeline.hpp	43
TimelineTrack	ECS	Source/Engine/ECS/AnimationTimeline.hp	56

		p	
TimelineTriggerSink	ECS	Source/Engine/ECS/AnimationTimeline.hpp	83
TimeScaleComponent	ECS	Source/Engine/ECS/Presentation.hpp	168
TrackKind	ECS	Source/Engine/ECS/AnimationTimeline.hpp	20
Transform2D	ECS	Source/Engine/ECS/Components.hpp	23
TransportGroupComponent	ECS	Source/Engine/ECS/Components.hpp	152
TransportMotion	ECS	Source/Engine/ECS/Components.hpp	138
TransportPath	ECS	Source/Engine/ECS/Components.hpp	129
TransportSurface	ECS	Source/Engine/ECS/Components.hpp	119
Vehicle2D	ECS	Source/Engine/ECS/Components.hpp	83
VehicleFlags	ECS	Source/Engine/ECS/Components.hpp	75
Velocity	ECS	Source/Engine/ECS/Components.hpp	186
View	ECS	Source/Engine/ECS/CommandLog.hpp	187
VirtualFileSystem	ECS	Source/Engine/ECS/Scene.hpp	18
VirtualFileSystem	ECS	Source/Engine/ECS/Schedule.hpp	20
VirtualFileSystem	ECS	Source/Engine/ECS/SkeletonAsset.hpp	18
VirtualFileSystem	ECS	Source/Engine/ECS/Tilemap.hpp	20
Wheel2D	ECS	Source/Engine/ECS/Components.hpp	99
WindowManager	ECS	Source/Engine/ECS/Schedule.hpp	19
WorldSnapshotV3	ECS	Source/Engine/ECS/SnapshotWorld.hpp	86

Engine/Gameplay (31)

Symbol	Kind	Header	Line
ActiveGameplayEffect	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	66
Entry	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	50
Entry	Gameplay	Source/Engine/Gameplay/GameplayTags.hpp	152
GameplayAttribute	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	21
GameplayAttributeModifier	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	82
GameplayAttributeRegistry	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	35
GameplayAttributesComponent	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	66
GameplayAttributeValues	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	60
GameplayEffectDef	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	45
GameplayEffectEvent	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	95
GameplayEffectEventType	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	88
GameplayEffectFlags	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	32
GameplayEffectsCo	Gameplay	Source/Engine/	78

mponent		Gameplay/ GameplayEffects.hpp	
GameplayEffectStackPolicy	Gameplay	Source/Engine/ Gameplay/ GameplayEffects.hpp	22
GameplayEffectSystem	Gameplay	Source/Engine/ Gameplay/ GameplayEffects.hpp	107
GameplayModifierOp	Gameplay	Source/Engine/ Gameplay/ GameplayAttributes.hpp	75
GameplaySkillDef	Gameplay	Source/Engine/ Gameplay/ GameplaySkills.hpp	55
GameplaySkillEvent	Gameplay	Source/Engine/ Gameplay/ GameplaySkills.hpp	98
GameplaySkillEventType	Gameplay	Source/Engine/ Gameplay/ GameplaySkills.hpp	33
GameplaySkillFailReason	Gameplay	Source/Engine/ Gameplay/ GameplaySkills.hpp	41
GameplaySkillsComponent	Gameplay	Source/Engine/ Gameplay/ GameplaySkills.hpp	89
GameplaySkillSlot	Gameplay	Source/Engine/ Gameplay/ GameplaySkills.hpp	77
GameplaySkillState	Gameplay	Source/Engine/ Gameplay/ GameplaySkills.hpp	27
GameplaySkillSystem	Gameplay	Source/Engine/ Gameplay/ GameplaySkills.hpp	110
GameplaySkillTargetKind	Gameplay	Source/Engine/ Gameplay/ GameplaySkills.hpp	19
GameplayTag	Gameplay	Source/Engine/ Gameplay/ GameplayTags.hpp	22
GameplayTagContainer	Gameplay	Source/Engine/ Gameplay/ GameplayTags.hpp	170

GameplayTagDatabase	Gameplay	Source/Engine/Gameplay/GameplayTagDatabase.hpp	18
GameplayTagMask	Gameplay	Source/Engine/Gameplay/GameplayTags.hpp	38
GameplayTagRegistry	Gameplay	Source/Engine/Gameplay/GameplayTags.hpp	132
GameplayTagsComponent	Gameplay	Source/Engine/Gameplay/GameplayTags.hpp	211

Engine/Input (14)

Symbol	Kind	Header	Line
InputActionDesc	Input	Source/Engine/Input/InputSystem.hpp	81
InputActionState	Input	Source/Engine/Input/InputSystem.hpp	108
InputActionType	Input	Source/Engine/Input/InputSystem.hpp	23
InputContextDesc	Input	Source/Engine/Input/InputSystem.hpp	87
InputEvent	Input	Source/Engine/Input/InputSystem.hpp	120
InputEventType	Input	Source/Engine/Input/InputSystem.hpp	60
InputMappingDesc	Input	Source/Engine/Input/InputSystem.hpp	94
InputModifier	Input	Source/Engine/Input/InputSystem.hpp	74
InputModifierKind	Input	Source/Engine/Input/InputSystem.hpp	39
InputSource	Input	Source/Engine/Input/InputSystem.hpp	68
InputSourceKind	Input	Source/Engine/Input/InputSystem.hpp	30
InputSystem	Input	Source/Engine/Input/InputSystem.hpp	132
InputTriggerKind	Input	Source/Engine/Input/InputSystem.hpp	50

UserSlot	Input	Source/Engine/Input/ InputSystem.hpp	175

Engine/Navigation (25)

Symbol	Kind	Header	Line
CrowdAgent	Navigation	Source/Engine/ Navigation/ CrowdPool.hpp	16
CrowdPool	Navigation	Source/Engine/ Navigation/ CrowdPool.hpp	46
CrowdRenderSettings	Navigation	Source/Engine/ Navigation/ CrowdRender.hpp	14
CrowdSettings	Navigation	Source/Engine/ Navigation/ CrowdPool.hpp	27
Entry	Navigation	Source/Engine/ Navigation/ FlowField.hpp	79
FlowField	Navigation	Source/Engine/ Navigation/ FlowField.hpp	33
FlowFieldDesc	Navigation	Source/Engine/ Navigation/ FlowField.hpp	22
NavAgentComponent	Navigation	Source/Engine/ Navigation/ Navigation.hpp	242
NavAgentPathSettings	Navigation	Source/Engine/ Navigation/ Navigation.hpp	206
NavAgentSettings	Navigation	Source/Engine/ Navigation/ Navigation.hpp	184
NavigationSystem	Navigation	Source/Engine/ Navigation/ Navigation.hpp	265
NavMap	Navigation	Source/Engine/ Navigation/ Navigation.hpp	27
NavMesh	Navigation	Source/Engine/ Navigation/	49

		NavMesh.hpp	
NavMesh	Navigation	Source/Engine/ Navigation/ Navigation.hpp	18
NavMeshPortal	Navigation	Source/Engine/ Navigation/ NavMesh.hpp	39
NavMeshRect	Navigation	Source/Engine/ Navigation/ NavMesh.hpp	19
NavPath	Navigation	Source/Engine/ Navigation/ Navigation.hpp	99
NavPathOptions	Navigation	Source/Engine/ Navigation/ Navigation.hpp	80
NavSmoothingMode	Navigation	Source/Engine/ Navigation/ Navigation.hpp	72
Page	Navigation	Source/Engine/ Navigation/ PagedNavGrid.hpp	105
PagedNavGrid	Navigation	Source/Engine/ Navigation/ PagedNavGrid.hpp	32
PagedNavGridDesc	Navigation	Source/Engine/ Navigation/ PagedNavGrid.hpp	23
PathFinder	Navigation	Source/Engine/ Navigation/ Navigation.hpp	127
PathStats	Navigation	Source/Engine/ Navigation/ Navigation.hpp	116
TilemapComponent	Navigation	Source/Engine/ Navigation/ PagedNavGrid.hpp	14

Engine/Network (89)

Symbol	Kind	Header	Line
CellCoordinate	Network	Source/Engine/ Network/ RelevanceGrid.hpp	168

ChecksumComparison	Network	Source/Engine/Network/NetRollback.hpp	457
Config	Network	Source/Engine/Network/NetRollback.hpp	316
DatagramSocket	Network	Source/Engine/Network/Transport.hpp	32
Endpoint	Network	Source/Engine/Network/NetLink.hpp	92
Frame	Network	Source/Engine/Network/NetLink.hpp	84
FrameInfo	Network	Source/Engine/Network/NetRollback.hpp	221
FrameMeta	Network	Source/Engine/Network/NetRollback.hpp	473
InputChannel	Network	Source/Engine/Network/NetProtocol.hpp	56
InputChannelStats	Network	Source/Engine/Network/NetProtocol.hpp	44
InputPacket	Network	Source/Engine/Network/NetProtocol.hpp	24
InputRingBuffer	Network	Source/Engine/Network/NetRollback.hpp	123
LobbyError	Network	Source/Engine/Network/Lobby.hpp	37
LobbyMessageType	Network	Source/Engine/Network/Lobby.hpp	25
LobbyRoomInfo	Network	Source/Engine/Network/Lobby.hpp	73
LobbyServer	Network	Source/Engine/Network/Lobby.hpp	93
LobbyServerConfig	Network	Source/Engine/Network/Lobby.hpp	82
LocalEntry	Network	Source/Engine/Network/NetRollback.hpp	205

Member	Network	Source/Engine/ Network/Lobby.hpp	126
MemoryEndpoint	Network	Source/Engine/ Network/NetLink.hpp	46
MemoryEndpoint	Network	Source/Engine/ Network/NetLink.hpp	113
MemoryNetwork	Network	Source/Engine/ Network/NetLink.hpp	53
NetAddress	Network	Source/Engine/ Network/ Transport.hpp	19
NetDiagnosticsSnapshot	Network	Source/Engine/ Network/ Diagnostics.hpp	12
NetDisconnectReason	Network	Source/Engine/ Network/Session.hpp	65
NetEntityId	Network	Source/Engine/ Network/ NetRollback.hpp	39
NetEntityMap	Network	Source/Engine/ Network/ NetRollback.hpp	167
NetEntry	Network	Source/Engine/ Network/ NetRollback.hpp	200
NetIdentity	Network	Source/Engine/ Network/Identity.hpp	32
NetInputFrame	Network	Source/Engine/ Network/ NetRollback.hpp	70
NetInputStatus	Network	Source/Engine/ Network/ NetRollback.hpp	99
NetLink	Network	Source/Engine/ Network/NetLink.hpp	19
NetLobbyClient	Network	Source/Engine/ Network/Lobby.hpp	160
NetLobbyClientConfig	Network	Source/Engine/ Network/Lobby.hpp	151
NetPlayerId	Network	Source/Engine/ Network/ NetRollback.hpp	59
NetPlayerSlot	Network	Source/Engine/	134

		Network/Session.hpp	
NetRole	Network	Source/Engine/ Network/Session.hpp	32
NetRollbackSession	Network	Source/Engine/ Network/ NetRollback.hpp	240
NetSession	Network	Source/Engine/ Network/Session.hpp	189
NetSessionCallbacks	Network	Source/Engine/ Network/Session.hpp	178
NetSessionConfig	Network	Source/Engine/ Network/Lobby.hpp	15
NetSessionConfig	Network	Source/Engine/ Network/Session.hpp	149
NetSessionMode	Network	Source/Engine/ Network/Session.hpp	22
NetSessionStatus	Network	Source/Engine/ Network/Session.hpp	46
NetSlotState	Network	Source/Engine/ Network/Session.hpp	58
NetTicket	Network	Source/Engine/ Network/Ticket.hpp	35
NetTickInputs	Network	Source/Engine/ Network/ NetRollback.hpp	110
NetTickMode	Network	Source/Engine/ Network/Session.hpp	40
NetTransport	Network	Source/Engine/ Network/ Transport.hpp	127
Outstanding	Network	Source/Engine/ Network/ Transport.hpp	100
Peer	Network	Source/Engine/ Network/ SecureSessionTransport.hpp	87
PendingEvent	Network	Source/Engine/ Network/Session.hpp	279
PendingState	Network	Source/Engine/ Network/ Replication.hpp	129
Phase	Network	Source/Engine/	256

		Network/ SecureChannel.hpp	
PhysicsSystem	Network	Source/Engine/ Network/ NetRollback.hpp	20
RateEntry	Network	Source/Engine/ Network/ SecureSessionTransport.hpp	120
Reassembly	Network	Source/Engine/ Network/ Transport.hpp	112
RelevanceBand	Network	Source/Engine/ Network/ RelevanceGrid.hpp	32
RelevanceCell	Network	Source/Engine/ Network/ RelevanceGrid.hpp	71
RelevanceConfig	Network	Source/Engine/ Network/ NetRollback.hpp	34
RelevanceConfig	Network	Source/Engine/ Network/ RelevanceGrid.hpp	56
RelevanceEntity	Network	Source/Engine/ Network/ RelevanceGrid.hpp	81
RelevanceGrid	Network	Source/Engine/ Network/ NetRollback.hpp	36
RelevanceGrid	Network	Source/Engine/ Network/ RelevanceGrid.hpp	131
RelevanceGridStats	Network	Source/Engine/ Network/ RelevanceGrid.hpp	121
RelevancePlan	Network	Source/Engine/ Network/ NetRollback.hpp	35
RelevancePlan	Network	Source/Engine/ Network/ RelevanceGrid.hpp	93
ReplicationStatus	Network	Source/Engine/ Network/ Replication.hpp	28

Result	Network	Source/Engine/ Network/ NetRollback.hpp	243
RollbackReport	Network	Source/Engine/ Network/ NetRollback.hpp	229
Room	Network	Source/Engine/ Network/Lobby.hpp	132
SecureChannel	Network	Source/Engine/ Network/ SecureChannel.hpp	198
SecureChannelConfig	Network	Source/Engine/ Network/ SecureChannel.hpp	151
SecureCookieIssuer	Network	Source/Engine/ Network/ SecureChannel.hpp	131
SecureEndpoint	Network	Source/Engine/ Network/ SecureEndpoint.hpp	16
SecureSessionAuthConfig	Network	Source/Engine/ Network/ SecureSessionTransport.hpp	24
SecureSessionTransport	Network	Source/Engine/ Network/ SecureSessionTransport.hpp	38
SecureStatus	Network	Source/Engine/ Network/ SecureChannel.hpp	65
Slot	Network	Source/Engine/ Network/ NetRollback.hpp	155
SnapshotReceiver	Network	Source/Engine/ Network/ Replication.hpp	177
SnapshotReceiverConfig	Network	Source/Engine/ Network/ Replication.hpp	166
SnapshotSender	Network	Source/Engine/ Network/ Replication.hpp	94
SnapshotSenderConfig	Network	Source/Engine/ Network/	70

		Replication.hpp	
SnapshotSenderStats	Network	Source/Engine/Network/Replication.hpp	80
SortableEntity	Network	Source/Engine/Network/RelevanceGrid.hpp	195
TicketStatus	Network	Source/Engine/Network/Ticket.hpp	54
TransportChannel	Network	Source/Engine/Network/Transport.hpp	63
TransportPeer	Network	Source/Engine/Network/Transport.hpp	85
TransportStats	Network	Source/Engine/Network/Transport.hpp	69

Engine/Physics (134)

Symbol	Kind	Header	Line
Body	Physics	Source/Engine/Physics/DodPhysicsBackend.hpp	57
BodyRecord	Physics	Source/Engine/Physics/Physics.hpp	284
BodyType	Physics	Source/Engine/Physics/Physics.hpp	44
ChainBreakEvent	Physics	Source/Engine/Physics/PhysicsJoints.hpp	314
ChainError	Physics	Source/Engine/Physics/PhysicsJoints.hpp	146
ChainRuntime	Physics	Source/Engine/Physics/PhysicsJoints.hpp	618
ClothComponent	Physics	Source/Engine/Physics/Cloth.hpp	31
ClothDesc	Physics	Source/Engine/Physics/Cloth.hpp	63
ClothFlags	Physics	Source/Engine/	20

		Physics/Cloth.hpp	
ClothHandle	Physics	Source/Engine/ Physics/Cloth.hpp	46
ClothInfo	Physics	Source/Engine/ Physics/Cloth.hpp	99
ClothRenderSystem	Physics	Source/Engine/ Physics/ ClothRenderSystem.hpp	15
ClothSystem	Physics	Source/Engine/ Physics/Cloth.hpp	110
Collider2D	Physics	Source/Engine/ Physics/Physics.hpp	81
CollisionFilterMatrix	Physics	Source/Engine/ Physics/ CollisionFilter.hpp	48
CollisionLayer	Physics	Source/Engine/ Physics/ CollisionFilter.hpp	15
ConnectionRuntime	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	590
DodPhysicsBackend	Physics	Source/Engine/ Physics/ DodPhysicsBackend.hpp	17
DodPhysicsBackend	Physics	Source/Engine/ Physics/Physics.hpp	20
FluidDesc	Physics	Source/Engine/ Physics/Fluid.hpp	114
FluidInteractorComponent	Physics	Source/Engine/ Physics/Fluid.hpp	208
FluidWaveSystem	Physics	Source/Engine/ Physics/Fluid.hpp	219
GerstnerField	Physics	Source/Engine/ Physics/Fluid.hpp	49
GerstnerSample	Physics	Source/Engine/ Physics/Fluid.hpp	39
GerstnerWave	Physics	Source/Engine/ Physics/Fluid.hpp	25
GravityBodyComponent	Physics	Source/Engine/ Physics/Gravity.hpp	77
GravityBodyMode	Physics	Source/Engine/	33

		Physics/Gravity.hpp	
GravityBodyShape	Physics	Source/Engine/ Physics/Gravity.hpp	39
GravityDirection	Physics	Source/Engine/ Physics/Gravity.hpp	15
GravityFalloff	Physics	Source/Engine/ Physics/Gravity.hpp	25
GravitySystem	Physics	Source/Engine/ Physics/Gravity.hpp	120
GridEntry	Physics	Source/Engine/ Physics/ DodPhysicsBackend.hpp	90
JointBreakReason	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	72
JointChainComponent	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	378
JointChainDesc	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	187
JointChainFlags	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	91
JointChainHandle	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	250
JointChainInfo	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	429
JointChainKind	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	54
JointChainSection	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	119
JointConnectionFlags	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	80
JointConnectionHandle	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	268
JointConnectionInfo	Physics	Source/Engine/	287

		Physics/ PhysicsJoints.hpp	
JointConnectionKind	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	64
JointConnectionParams	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	559
JointLinkComponent	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	33
JointLinkFlags	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	22
JointTieDesc	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	328
JointTieSave	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	351
LevelGravityComponent	Physics	Source/Engine/ Physics/Gravity.hpp	53
ParticleEmitterComponent	Physics	Source/Engine/ Physics/Particles.hpp	30
ParticleEmitterFlags	Physics	Source/Engine/ Physics/Particles.hpp	14
ParticleEmitterGpu	Physics	Source/Engine/ Physics/Particles.hpp	65
ParticleSeedPush	Physics	Source/Engine/ Physics/Particles.hpp	88
ParticleStateGpu	Physics	Source/Engine/ Physics/Particles.hpp	96
PhysicsBackendKind	Physics	Source/Engine/ Physics/Physics.hpp	27
PhysicsBodyRemapEntry	Physics	Source/Engine/ Physics/ SnapshotRemap.hpp	17
PhysicsBodyRemapTable	Physics	Source/Engine/ Physics/ SnapshotRemap.hpp	24
PhysicsCapabilities	Physics	Source/Engine/ Physics/Physics.hpp	34

PhysicsContactEvent	Physics	Source/Engine/ Physics/Physics.hpp	109
PhysicsDesc	Physics	Source/Engine/ Physics/Physics.hpp	145
PhysicsEventType	Physics	Source/Engine/ Physics/Physics.hpp	102
PhysicsJointSystem	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	444
PhysicsRayHit	Physics	Source/Engine/ Physics/Physics.hpp	120
PhysicsSystem	Physics	Source/Engine/ Physics/Physics.hpp	159
PhysicsSystem	Physics	Source/Engine/ Physics/ TransportGroups.hpp	13
Projectile2D	Physics	Source/Engine/ Physics/Physics.hpp	134
ProjectileHitEvent	Physics	Source/Engine/ Physics/ ProjectileSystem.hpp	12
ProjectileSystem	Physics	Source/Engine/ Physics/ ProjectileSystem.hpp	24
RigidBody2D	Physics	Source/Engine/ Physics/Physics.hpp	64
RigidBodyFlags	Physics	Source/Engine/ Physics/Physics.hpp	51
Runtime	Physics	Source/Engine/ Physics/Cloth.hpp	156
SdfAtlasImage	Physics	Source/Engine/ Physics/SdfAtlas.hpp	18
SdfBrickGpu	Physics	Source/Engine/ Physics/ SdfBrickMap.hpp	48
SdfBrickHeader	Physics	Source/Engine/ Physics/ SdfBrickMap.hpp	32
SdfBrickHeader	Physics	Source/Engine/ Physics/ SdfCollision.hpp	103
SdfBrickMap	Physics	Source/Engine/ Physics/	79

		SdfBrickMap.hpp	
SdfBrickMap	Physics	Source/Engine/ Physics/ SdfCollision.hpp	102
SdfBrickMapDesc	Physics	Source/Engine/ Physics/ SdfBrickMap.hpp	63
SdfBrickState	Physics	Source/Engine/ Physics/ SdfBrickMap.hpp	24
SdfCarveOp	Physics	Source/Engine/ Physics/ SdfDestructible.hpp	25
SdfCarveReplayContext	Physics	Source/Engine/ Physics/ SdfDestructionCommands.hpp	37
SdfCarveShape	Physics	Source/Engine/ Physics/ SdfDestructible.hpp	17
SdfCollisionDesc	Physics	Source/Engine/ Physics/ SdfCollision.hpp	45
SdfCollisionSystem	Physics	Source/Engine/ Physics/ SdfCollision.hpp	105
SdfComposeLayer	Physics	Source/Engine/ Physics/ SdfComposer.hpp	25
SdfComposeOp	Physics	Source/Engine/ Physics/ SdfComposer.hpp	16
SdfComposer	Physics	Source/Engine/ Physics/ SdfComposer.hpp	37
SdfComposerSession	Physics	Source/Engine/ Physics/ SdfComposerSession.hpp	15
SdfConnection	Physics	Source/Engine/ Physics/ SdfDamage.hpp	66
SdfContact	Physics	Source/Engine/ Physics/ SdfCollision.hpp	62

SdfContactManifold	Physics	Source/Engine/ Physics/ SdfCollision.hpp	93
SdfContactPoint	Physics	Source/Engine/ Physics/ SdfCollision.hpp	82
SdfDamageModel	Physics	Source/Engine/ Physics/ SdfDamage.hpp	188
SdfDebrisPiece	Physics	Source/Engine/ Physics/SdfDebris.hpp	19
SdfDebrisSettings	Physics	Source/Engine/ Physics/SdfDebris.hpp	30
SdfDestructible	Physics	Source/Engine/ Physics/ SdfDestructible.hpp	56
SdfDestructible	Physics	Source/Engine/ Physics/SdfWorld.hpp	16
SdfDestructibleDesc	Physics	Source/Engine/ Physics/ SdfDestructible.hpp	36
SdfField	Physics	Source/Engine/ Physics/ SdfCollision.hpp	19
SdfGiAsset	Physics	Source/Engine/ Physics/ SdfGiAsset.hpp	65
SdfGiAssetHeader	Physics	Source/Engine/ Physics/ SdfGiAsset.hpp	45
SdfGiBaked	Physics	Source/Engine/ Physics/ SdfGiBake.hpp	44
SdfGiBakedProbe	Physics	Source/Engine/ Physics/ SdfGiBake.hpp	32
SdfGiBakeSettings	Physics	Source/Engine/ Physics/ SdfGiBake.hpp	16
SdfGiProbe	Physics	Source/Engine/ Physics/SdfGi.hpp	21
SdfGiVolume	Physics	Source/Engine/ Physics/SdfGi.hpp	40
SdfImpact	Physics	Source/Engine/	85

		Physics/ SdfDamage.hpp	
SdfImpactResult	Physics	Source/Engine/ Physics/ SdfDamage.hpp	100
SdfMaterialProfile	Physics	Source/Engine/ Physics/ SdfDamage.hpp	25
SdfPart	Physics	Source/Engine/ Physics/ SdfDamage.hpp	47
SdfPrimitiveInfo	Physics	Source/Engine/ Physics/ SdfPrimitives.hpp	137
SdfShape	Physics	Source/Engine/ Physics/ SdfPrimitives.hpp	82
SdfShapeComponent	Physics	Source/Engine/ Physics/ SdfPrimitives.hpp	127
SdfShapeType	Physics	Source/Engine/ Physics/ SdfPrimitives.hpp	16
SdfSmokeDesc	Physics	Source/Engine/ Physics/ SdfSmoke.hpp	15
SdfSmokeField	Physics	Source/Engine/ Physics/ SdfSmoke.hpp	32
SdfVolumeAsset	Physics	Source/Engine/ Physics/ SdfVolumeAsset.hpp	52
SdfVolumeHeader	Physics	Source/Engine/ Physics/ SdfVolumeAsset.hpp	26
SdfWorld	Physics	Source/Engine/ Physics/SdfWorld.hpp	60
SdfWorldBox	Physics	Source/Engine/ Physics/SdfWorld.hpp	45
SdfWorldCircle	Physics	Source/Engine/ Physics/SdfWorld.hpp	51
SdfWorldDesc	Physics	Source/Engine/ Physics/SdfWorld.hpp	31
SdfWorldImprint	Physics	Source/Engine/	20

		Physics/SdfWorld.hpp	
ShallowWaterGrid	Physics	Source/Engine/ Physics/Fluid.hpp	134
Storage	Physics	Source/Engine/ Physics/Cloth.hpp	155
Storage	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	642
TilemapComponent	Physics	Source/Engine/ Physics/SdfWorld.hpp	15
Tracked	Physics	Source/Engine/ Physics/Fluid.hpp	248
TransportCapabilities	Physics	Source/Engine/ Physics/ TransportGroups.hpp	31
TransportGroupState	Physics	Source/Engine/ Physics/ TransportGroups.hpp	41
TransportGroupSystem	Physics	Source/Engine/ Physics/ TransportGroups.hpp	63
WaterView	Physics	Source/Engine/ Physics/Fluid.hpp	85
WheelSystem	Physics	Source/Engine/ Physics/ VehicleWheels.hpp	16

Engine/Platform (21)

Symbol	Kind	Header	Line
AchievementFlags	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	95
AchievementRequest	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	104
CloudSaveComponent	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	124
CloudSaveFlags	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	115
EgsStoreSystem	Platform	Source/Engine/ Platform/	178

		PlatformSystems.hpp	
GogAchievementSystem	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	224
GogStorageSystem	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	258
PlatformAccountComponent	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	137
PlatformCommand	Platform	Source/Engine/ Platform/ PlatformService.hpp	35
PlatformCommandLog	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	46
PlatformCommandType	Platform	Source/Engine/ Platform/ PlatformService.hpp	24
PlatformLogEvent	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	35
PlatformLogType	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	26
PlatformService	Platform	Source/Engine/ Platform/ PlatformService.hpp	76
PlatformServiceCallbacks	Platform	Source/Engine/ Platform/ PlatformService.hpp	51
PlatformServiceDesc	Platform	Source/Engine/ Platform/ PlatformService.hpp	71
PlatformServiceStats	Platform	Source/Engine/ Platform/ PlatformService.hpp	59
PlatformState	Platform	Source/Engine/ Platform/ PlatformService.hpp	132
StagedSave	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	204
StagedSave	Platform	Source/Engine/ Platform/	291

		PlatformSystems.hpp	
SteamAchievementSystem	Platform	Source/Engine/Platform/PlatformSystems.hpp	148

Engine/RHI (161)

Symbol	Kind	Header	Line
Api	RHI	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	122
Backend	RHI	Source/Engine/RHI/RHI.hpp	19
BackendCapabilities	RHI	Source/Engine/RHI/UpscalerCapabilities.hpp	28
BackendSelection	RHI	Source/Engine/RHI/UpscalerBackend.hpp	28
CameraData	RHI	Source/Engine/RHI/RenderItem.hpp	240
CameraDistortionFlags	RHI	Source/Engine/RHI/CameraStack.hpp	30
CameraDistortionSettings	RHI	Source/Engine/RHI/CameraStack.hpp	40
CameraStackDesc	RHI	Source/Engine/RHI/CameraStack.hpp	56
CameraStackFlags	RHI	Source/Engine/RHI/CameraStack.hpp	17
CameraStackSystem	RHI	Source/Engine/RHI/CameraStack.hpp	81
CameraStackSystem	RHI	Source/Engine/RHI/RenderGraph.hpp	16
CameraStackSystem	RHI	Source/Engine/RHI/RenderPasses.hpp	12
ChangeMaskRenderExtract	RHI	Source/Engine/RHI/ChangeMaskExtract.hpp	33
ClusterGridConstants	RHI	Source/Engine/RHI/GpuLighting.hpp	46
ClusterHeader	RHI	Source/Engine/RHI/GpuLighting.hpp	36
ColorGradingConsta	RHI	Source/Engine/RHI/	128

nts		RenderPasses.hpp	
ColorGradingPass	RHI	Source/Engine/RHI/RenderPasses.hpp	136
ColorGradingSettings	RHI	Source/Engine/RHI/RenderPasses.hpp	116
CookedMaterialView	RHI	Source/Engine/RHI/MaterialAsset.hpp	152
CustomFrameGenerator	RHI	Source/Engine/RHI/CustomFrameGenerator.hpp	10
DeviceDesc	RHI	Source/Engine/RHI/RHI.hpp	49
DeviceInfo	RHI	Source/Engine/RHI/RHI.hpp	112
DirtyRange	RHI	Source/Engine/RHI/RenderItem.hpp	221
DlssAdapter	RHI	Source/Engine/RHI/DlssAdapter.hpp	33
DlssQualityMode	RHI	Source/Engine/RHI/DlssAdapter.hpp	9
DlssRuntimeAvailability	RHI	Source/Engine/RHI/DlssRuntime.hpp	9
DlssRuntimeProbe	RHI	Source/Engine/RHI/DlssRuntimeProbe.hpp	14
DrawBatch	RHI	Source/Engine/RHI/RenderItem.hpp	201
DrawIndexedIndirectCommand	RHI	Source/Engine/RHI/GpuVisibility.hpp	15
DynamicResolution	RHI	Source/Engine/RHI/DynamicResolution.hpp	19
DynamicResolutionSettings	RHI	Source/Engine/RHI/DynamicResolution.hpp	9
Entry	RHI	Source/Engine/RHI/MaterialPipelineCache.hpp	74
ExportStatus	RHI	Source/Engine/RHI/FrameExporter.hpp	100
Extent2D	RHI	Source/Engine/RHI/RHI.hpp	40

ExtractStats	RHI	Source/Engine/RHI/ChangeMaskExtract.hpp	21
FidelityFxRuntimeAvailability	RHI	Source/Engine/RHI/FidelityFxRuntime.hpp	25
FrameAccessibility	RHI	Source/Engine/RHI/RenderItem.hpp	262
FrameAccessibilityFlags	RHI	Source/Engine/RHI/RenderItem.hpp	249
FrameExportDesc	RHI	Source/Engine/RHI/FrameExporter.hpp	53
FrameExporter	RHI	Source/Engine/RHI/FrameExporter.hpp	112
FrameGenBackend	RHI	Source/Engine/RHI/UpscalerBackend.hpp	19
FrameGenerationInput	RHI	Source/Engine/RHI/FrameGeneration.hpp	10
FrameGenerationPolicy	RHI	Source/Engine/RHI/FrameGenerationPolicy.hpp	19
FrameGenerationStatus	RHI	Source/Engine/RHI/FrameGenerationPolicy.hpp	9
FramePacingConfig	RHI	Source/Engine/RHI/FramePacing.hpp	10
FramePacingState	RHI	Source/Engine/RHI/FramePacing.hpp	17
FramePacket	RHI	Source/Engine/RHI/RenderItem.hpp	372
FrameSink	RHI	Source/Engine/RHI/FrameExporter.hpp	85
FrameStats	RHI	Source/Engine/RHI/RHI.hpp	257
Fsr2Adapter	RHI	Source/Engine/RHI/Fsr2Adapter.hpp	33
Fsr2Pass	RHI	Source/Engine/RHI/Fsr2Pass.hpp	10
Fsr2QualityMode	RHI	Source/Engine/RHI/Fsr2Adapter.hpp	9
Fsr3FrameGenerator	RHI	Source/Engine/RHI/Fsr3FrameGenerator.hpp	10

GpuCullingConstants	RHI	Source/Engine/RHI/GpuVisibility.hpp	28
GpuGiProbe	RHI	Source/Engine/RHI/GpuLighting.hpp	67
GpuGiVolumeHeader	RHI	Source/Engine/RHI/GpuLighting.hpp	57
GpuLight	RHI	Source/Engine/RHI/GpuLighting.hpp	18
GpuVendor	RHI	Source/Engine/RHI/UpscalerCapabilities.hpp	19
HashEntry	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	197
IFrameGenerator	RHI	Source/Engine/RHI/FrameGeneration.hpp	27
ImageLayout	RHI	Source/Engine/RHI/RHI.hpp	218
ITemporalUpscaler	RHI	Source/Engine/RHI/TemporalUpscaler.hpp	28
KeySorter	RHI	Source/Engine/RHI/Sort.hpp	14
LookupEntry	RHI	Source/Engine/RHI/ChangeMaskExtract.hpp	255
MaterialAssetError	RHI	Source/Engine/RHI/MaterialAsset.hpp	42
MaterialCompileOptions	RHI	Source/Engine/RHI/MaterialGraphCompiler.hpp	27
MaterialContract	RHI	Source/Engine/RHI/MaterialAsset.hpp	83
MaterialDomain	RHI	Source/Engine/RHI/MaterialGraphCompiler.hpp	21
MaterialGraph	RHI	Source/Engine/RHI/MaterialGraph.hpp	555
MaterialGraphCompiler	RHI	Source/Engine/RHI/MaterialGraphCompiler.hpp	58
MaterialGraphRequirements	RHI	Source/Engine/RHI/MaterialGraph.hpp	542

MaterialNode	RHI	Source/Engine/RHI/MaterialGraph.hpp	117
MaterialNodeFlags	RHI	Source/Engine/RHI/MaterialGraph.hpp	111
MaterialNodeKind	RHI	Source/Engine/RHI/MaterialGraph.hpp	29
MaterialParam	RHI	Source/Engine/RHI/MaterialGraph.hpp	99
MaterialParamKind	RHI	Source/Engine/RHI/MaterialGraph.hpp	88
MaterialPin	RHI	Source/Engine/RHI/MaterialGraph.hpp	129
MaterialPipelineCache	RHI	Source/Engine/RHI/MaterialPipelineCache.hpp	38
MaterialPipelineCacheStats	RHI	Source/Engine/RHI/MaterialPipelineCache.hpp	18
MaterialShaderProgram	RHI	Source/Engine/RHI/MaterialGraphCompiler.hpp	39
MaterialValueType	RHI	Source/Engine/RHI/MaterialGraph.hpp	78
MeshDesc	RHI	Source/Engine/RHI/Mesh.hpp	102
MeshDrawData	RHI	Source/Engine/RHI/Mesh.hpp	70
MeshDrawFlags	RHI	Source/Engine/RHI/Mesh.hpp	54
MeshHandle	RHI	Source/Engine/RHI/Mesh.hpp	87
MeshItemFlags	RHI	Source/Engine/RHI/Mesh.hpp	62
MeshRenderItem	RHI	Source/Engine/RHI/Mesh.hpp	114
MeshSkinRange	RHI	Source/Engine/RHI/Mesh.hpp	46
MeshVertex	RHI	Source/Engine/RHI/Mesh.hpp	17
MotionVectorItem	RHI	Source/Engine/RHI/RenderItem.hpp	354
NativeDeviceHandle	RHI	Source/Engine/RHI/	708

s		RHI.hpp	
NativeTextureView	RHI	Source/Engine/RHI/RHI.hpp	224
OitPass	RHI	Source/Engine/RHI/OitPass.hpp	11
PageSlot	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	188
PlatformKind	RHI	Source/Engine/RHI/UpscalerCapabilities.hpp	11
PostFxConstants	RHI	Source/Engine/RHI/RenderPasses.hpp	97
PostFxFlags	RHI	Source/Engine/RHI/PostProcess.hpp	10
PostFxPass	RHI	Source/Engine/RHI/RenderPasses.hpp	157
PostFxSettings	RHI	Source/Engine/RHI/PostProcess.hpp	30
RenderBlendMode	RHI	Source/Engine/RHI/RHI.hpp	190
RenderCommandRecord	RHI	Source/Engine/RHI/RenderGraph.hpp	52
RenderFlags	RHI	Source/Engine/RHI/RenderItem.hpp	42
RenderGraph	RHI	Source/Engine/RHI/RenderGraph.hpp	119
RenderGraphContext	RHI	Source/Engine/RHI/RenderGraph.hpp	69
RenderGraphPass	RHI	Source/Engine/RHI/RenderGraph.hpp	106
RenderItem	RHI	Source/Engine/RHI/RenderItem.hpp	90
RenderPassDesc	RHI	Source/Engine/RHI/RenderGraph.hpp	94
RenderPassKind	RHI	Source/Engine/RHI/RenderGraph.hpp	26
RenderResourceKind	RHI	Source/Engine/RHI/RenderGraph.hpp	33
RenderTargetDesc	RHI	Source/Engine/RHI/RHI.hpp	197
RenderTargetHandle	RHI	Source/Engine/RHI/	207

		RHI.hpp	
RHI	RHI	Source/Engine/RHI/RHI.hpp	328
SdfLightingFrame	RHI	Source/Engine/RHI/RenderItem.hpp	310
SdfLightingOccluder	RHI	Source/Engine/RHI/RenderItem.hpp	302
SdfShadowConstants	RHI	Source/Engine/RHI/RenderPasses.hpp	37
SdfShadowDesc	RHI	Source/Engine/RHI/RenderPasses.hpp	18
SdfShadowPass	RHI	Source/Engine/RHI/RenderPasses.hpp	66
SequenceFooter	RHI	Source/Engine/RHI/FrameExporter.hpp	42
SequenceHeader	RHI	Source/Engine/RHI/FrameExporter.hpp	31
ShaderPipelineDesc	RHI	Source/Engine/RHI/RHI.hpp	167
ShaderPipelineDomain	RHI	Source/Engine/RHI/RHI.hpp	161
ShaderPipelineHandle	RHI	Source/Engine/RHI/RHI.hpp	178
SkinnedVertex	RHI	Source/Engine/RHI/Mesh.hpp	30
Slot	RHI	Source/Engine/RHI/CameraStack.hpp	142
SpatialUpscaleMode	RHI	Source/Engine/RHI/RHI.hpp	90
StaticBakeCache	RHI	Source/Engine/RHI/StaticBake.hpp	58
StaticBakeRenderer	RHI	Source/Engine/RHI/StaticBakeRenderer.hpp	14
StaticBakeSettings	RHI	Source/Engine/RHI/StaticBake.hpp	10
StaticBakeStats	RHI	Source/Engine/RHI/StaticBake.hpp	47
StaticBakeTileKey	RHI	Source/Engine/RHI/StaticBake.hpp	17
TemporalResolvePass	RHI	Source/Engine/RHI/TemporalResolvePass	11

		.hpp	
TemporalUpscaleInput	RHI	Source/Engine/RHI/TemporalUpscaler.hpp	11
TextureArrayDesc	RHI	Source/Engine/RHI/RHI.hpp	146
TextureArrayHandle	RHI	Source/Engine/RHI/RHI.hpp	304
Tile	RHI	Source/Engine/RHI/StaticBake.hpp	90
TintEntry	RHI	Source/Engine/RHI/ViewportPresenter.hpp	81
TransientRing	RHI	Source/Engine/RHI/RingBuffer.hpp	15
UiBackendInfo	RHI	Source/Engine/RHI/RHI.hpp	234
UpscalerBackend	RHI	Source/Engine/RHI/UpscalerBackend.hpp	9
UpscalerCaps	RHI	Source/Engine/RHI/RHI.hpp	98
ViewportMode	RHI	Source/Engine/RHI/ViewportModes.hpp	15
ViewportModeDescriptor	RHI	Source/Engine/RHI/ViewportModes.hpp	46
ViewportModeLegend	RHI	Source/Engine/RHI/ViewportModes.hpp	37
ViewportPresenter	RHI	Source/Engine/RHI/ViewportPresenter.hpp	29
ViewportPresenterStats	RHI	Source/Engine/RHI/ViewportPresenter.hpp	17
VirtualFileSystem	RHI	Source/Engine/RHI/FrameExporter.hpp	15
VirtualFileSystem	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	16
VirtualTextureAddress	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	68
VirtualTextureCamera	RHI	Source/Engine/RHI/VirtualTextureSystem.	59

		hpp	
VirtualTextureDesc	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	33
VirtualTextureHandle	RHI	Source/Engine/RHI/RenderItem.hpp	279
VirtualTexturePageLocation	RHI	Source/Engine/RHI/VirtualTextureGpu.hpp	11
VirtualTextureResourceDesc	RHI	Source/Engine/RHI/RHI.hpp	316
VirtualTextureRhiTarget	RHI	Source/Engine/RHI/VirtualTextureGpu.hpp	118
VirtualTextureSample	RHI	Source/Engine/RHI/VirtualTextureGpu.hpp	58
VirtualTextureStats	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	45
VirtualTextureSystem	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	111
VirtualTextureTableUpdate	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	101
VirtualTextureTileId	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	92
VirtualTextureTileUpload	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	77
VisibleSprite	RHI	Source/Engine/RHI/ViewportPresenter.hpp	87

Engine/Script (67)

Symbol	Kind	Header	Line
AotBuilder	Script	Source/Engine/Script/AotBuilder.hpp	71
AotBuildOptions	Script	Source/Engine/Script/AotBuilder.hpp	43
AotBuildResult	Script	Source/Engine/Script/AotBuilder.hpp	60
AotBuildState	Script	Source/Engine/	19

		Script/AotBuilder.hpp	
AotGraphTranspileOptions	Script	Source/Engine/Script/AotGraphCompiler.hpp	13
AotHost	Script	Source/Engine/Script/AotHost.hpp	32
AotHost	Script	Source/Engine/Script/ScriptSystem.hpp	23
AotModule	Script	Source/Engine/Script/AotModule.hpp	22
AotProgramRef	Script	Source/Engine/Script/AotHost.hpp	22
AotTranspileOptions	Script	Source/Engine/Script/AotTranspiler.hpp	12
BoundProgram	Script	Source/Engine/Script/EcsBridge.hpp	101
CameraStackSystem	Script	Source/Engine/Script/ScriptSystem.hpp	18
CameraStackSystem	Script	Source/Engine/Script/Vm.hpp	18
Code	Script	Source/Engine/Script/Vm.hpp	32
CompiledDebug	Script	Source/Engine/Script/Bytecode.hpp	488
CompiledProgram	Script	Source/Engine/Script/Bytecode.hpp	503
ComponentAccessor	Script	Source/Engine/Script/EcsBridge.hpp	17
ComponentBinding	Script	Source/Engine/Script/Bytecode.hpp	432
Context	Script	Source/Engine/Script/AotAbi.hpp	112
DebugField	Script	Source/Engine/Script/ScriptDebug.hpp	123
DebugFrame	Script	Source/Engine/Script/ScriptDebug.hpp	103
DebugSectionHeade	Script	Source/Engine/	181

r		Script/Bytecode.hpp	
DebugSpan	Script	Source/Engine/ Script/Bytecode.hpp	159
DebugSpanFlags	Script	Source/Engine/ Script/Bytecode.hpp	150
DebugVariable	Script	Source/Engine/ Script/Bytecode.hpp	172
EcsBridge	Script	Source/Engine/ Script/EcsBridge.hpp	26
FieldBinding	Script	Source/Engine/ Script/Bytecode.hpp	442
FunctionEntry	Script	Source/Engine/ Script/Bytecode.hpp	455
HostApi	Script	Source/Engine/ Script/AotAbi.hpp	54
Instruction	Script	Source/Engine/ Script/Bytecode.hpp	412
LogicCompareOp	Script	Source/Engine/ Script/ GraphCompiler.hpp	22
LogicCompileOptions	Script	Source/Engine/ Script/ GraphCompiler.hpp	741
LogicCompileResult	Script	Source/Engine/ Script/ GraphCompiler.hpp	748
LogicFieldBinding	Script	Source/Engine/ Script/ GraphCompiler.hpp	90
LogicGraph	Script	Source/Engine/ Script/ GraphCompiler.hpp	350
LogicLink	Script	Source/Engine/ Script/ GraphCompiler.hpp	339
LogicMathOp	Script	Source/Engine/ Script/ GraphCompiler.hpp	56
LogicNode	Script	Source/Engine/ Script/ GraphCompiler.hpp	321
LogicNodeKind	Script	Source/Engine/ Script/	111

		GraphCompiler.hpp	
Module	Script	Source/Engine/ Script/AotAbi.hpp	379
NativeId	Script	Source/Engine/ Script/Bytecode.hpp	18
OpCode	Script	Source/Engine/ Script/Bytecode.hpp	230
PendingField	Script	Source/Engine/ Script/Bytecode.hpp	569
PhysicsSystem	Script	Source/Engine/ Script/ ScriptSystem.hpp	17
PhysicsSystem	Script	Source/Engine/ Script/Vm.hpp	17
PinType	Script	Source/Engine/ Script/ GraphCompiler.hpp	306
ProgramBuilder	Script	Source/Engine/ Script/Bytecode.hpp	534
ProgramHeader	Script	Source/Engine/ Script/Bytecode.hpp	465
ResolvedComponent	Script	Source/Engine/ Script/EcsBridge.hpp	93
ScriptComponent	Script	Source/Engine/ Script/ ScriptComponent.hpp	16
ScriptEvent	Script	Source/Engine/ Script/Bytecode.hpp	205
ScriptRuntime	Script	Source/Engine/ Script/ ScriptSystem.hpp	22
ScriptRuntime	Script	Source/Engine/ Script/ ScriptSystem.hpp	26
ScriptVm	Script	Source/Engine/ Script/Vm.hpp	115
ScriptVm	Script	Source/Engine/ Script/Vm.hpp	229
Slot	Script	Source/Engine/ Script/AotHost.hpp	94
Slot	Script	Source/Engine/ Script/ ScriptSystem.hpp	140

SourceLocation	Script	Source/Engine/ Script/ ScriptDebug.hpp	17
TextCompileOptions	Script	Source/Engine/ Script/TextScript.hpp	14
TransportGroupSyst em	Script	Source/Engine/ Script/Vm.hpp	16
Vector2D	Script	Source/Engine/ Script/AotAbi.hpp	28
VmContext	Script	Source/Engine/ Script/Vm.hpp	94
VmDebugState	Script	Source/Engine/ Script/Vm.hpp	154
VmFault	Script	Source/Engine/ Script/Vm.hpp	30
VmHalt	Script	Source/Engine/ Script/Vm.hpp	127
VmPauseReason	Script	Source/Engine/ Script/Vm.hpp	118
VmStopInfo	Script	Source/Engine/ Script/Vm.hpp	134

Engine/Text (68)

Symbol	Kind	Header	Line
AssetSlot	Text	Source/Engine/Text/ TextStore.hpp	218
ChainedShaper	Text	Source/Engine/Text/ TextShaper.hpp	133
CodepointShaper	Text	Source/Engine/Text/ TextShaper.hpp	99
Entry	Text	Source/Engine/Text/ TextShaper.hpp	247
FontCandidate	Text	Source/Engine/Text/ TextFontStack.hpp	24
FontSlot	Text	Source/Engine/Text/ TextFontStack.hpp	59
FormatEntry	Text	Source/Engine/Text/ TextService.hpp	583
FormattedLayoutEnt ry	Text	Source/Engine/Text/ TextService.hpp	566
FormattedLayoutKey	Text	Source/Engine/Text/	556

		TextService.hpp	
HarfBuzzShaper	Text	Source/Engine/Text/TextShaper.hpp	110
Impl	Text	Source/Engine/Text/TextShaper.hpp	128
ITextShaper	Text	Source/Engine/Text/TextShaper.hpp	89
LayoutEntry	Text	Source/Engine/Text/TextService.hpp	548
LayoutKey	Text	Source/Engine/Text/TextService.hpp	528
MappedTextAsset	Text	Source/Engine/Text/TextAssetFile.hpp	17
MarkupStackEntry	Text	Source/Engine/Text/TextMarkup.hpp	71
MarkupVariantEntry	Text	Source/Engine/Text/TextService.hpp	574
ScriptBackend	Text	Source/Engine/Text/TextShapeSystem.hpp	111
ScriptShaper	Text	Source/Engine/Text/TextShaper.hpp	207
ShapedGlyph	Text	Source/Engine/Text/TextShaper.hpp	56
ShapedGlyphHit	Text	Source/Engine/Text/TextShapedLayout.hpp	153
ShapedLayoutResult	Text	Source/Engine/Text/TextShapedLayout.hpp	26
ShapedLine	Text	Source/Engine/Text/TextShapedLayout.hpp	18
StyledAsset	Text	Source/Engine/Text/TextService.hpp	595
TextAction	Text	Source/Engine/Text/TextStyle.hpp	61
TextActionKind	Text	Source/Engine/Text/TextStyle.hpp	51
TextAnimationParams	Text	Source/Engine/Text/TextRender.hpp	29
TextAssetBlock	Text	Source/Engine/Text/	34

		TextAssetFormat.hpp	
TextAssetHeader	Text	Source/Engine/Text/ TextAssetFormat.hpp	20
TextAssetRef	Text	Source/Engine/Text/ TextAsset.hpp	14
TextAssetStatus	Text	Source/Engine/Text/ TextAssetFormat.hpp	63
TextAssetView	Text	Source/Engine/Text/ TextAssetFormat.hpp	42
TextBlock	Text	Source/Engine/Text/ TextAsset.hpp	37
TextDirection	Text	Source/Engine/Text/ TextShaper.hpp	21
TextFontStack	Text	Source/Engine/Text/ TextFontStack.hpp	54
TextFormatArg	Text	Source/Engine/Text/ TextFormat.hpp	45
TextFormatStatus	Text	Source/Engine/Text/ TextFormat.hpp	33
TextGradientParams	Text	Source/Engine/Text/ TextRender.hpp	22
TextInlineHit	Text	Source/Engine/Text/ TextLayout.hpp	204
TextInlineKind	Text	Source/Engine/Text/ TextStyle.hpp	73
TextInlineObject	Text	Source/Engine/Text/ TextStyle.hpp	83
TextInlineSprite	Text	Source/Engine/Text/ TextStyle.hpp	99
TextLayoutParams	Text	Source/Engine/Text/ TextLayout.hpp	54
TextLayoutResult	Text	Source/Engine/Text/ TextLayout.hpp	72
TextLine	Text	Source/Engine/Text/ TextLayout.hpp	22
TextMarkupStatus	Text	Source/Engine/Text/ TextMarkup.hpp	16
TextRenderEffects	Text	Source/Engine/Text/ TextRender.hpp	44
TextRenderParams	Text	Source/Engine/Text/ TextRender.hpp	57

TextRenderStats	Text	Source/Engine/Text/TextRender.hpp	76
TextRenderSystemStats	Text	Source/Engine/Text/TextRenderSystem.hpp	19
TextRun	Text	Source/Engine/Text/TextShaper.hpp	47
TextRuntime	Text	Source/Engine/Text/TextRuntime.hpp	17
TextScript	Text	Source/Engine/Text/TextShaper.hpp	30
TextService	Text	Source/Engine/Text/TextService.hpp	18
TextShapeCache	Text	Source/Engine/Text/TextShapeCache.hpp	59
TextShapeHandle	Text	Source/Engine/Text/TextShapeCache.hpp	19
TextShapeKey	Text	Source/Engine/Text/TextShapeCache.hpp	32
TextShapeParams	Text	Source/Engine/Text/TextShaper.hpp	71
TextShapeResult	Text	Source/Engine/Text/TextShaper.hpp	82
TextShapeSystemStats	Text	Source/Engine/Text/TextShapeSystem.hpp	16
TextShapeView	Text	Source/Engine/Text/TextShapeCache.hpp	50
TextSpan	Text	Source/Engine/Text/TextStyle.hpp	40
TextStore	Text	Source/Engine/Text/TextStore.hpp	19
TextStyle	Text	Source/Engine/Text/TextStyle.hpp	27
TextStyledBlock	Text	Source/Engine/Text/TextMarkup.hpp	30
TextStyleFlags	Text	Source/Engine/Text/TextStyle.hpp	12
UnicodeBoundary	Text	Source/Engine/Text/UnicodeBreak.hpp	13
VerticalFormEntry	Text	Source/Engine/Text/TextVertical.hpp	16

Engine/UI (101)

Symbol	Kind	Header	Line
AccessibilityEditorSettings	UI	Source/Engine/UI/Accessibility.hpp	250
AccessibilityFlags	UI	Source/Engine/UI/Accessibility.hpp	208
AccessibilityKernel	UI	Source/Engine/UI/Accessibility.hpp	304
AccessibilityOverlay	UI	Source/Engine/UI/AccessibilityOverlay.hpp	30
AccessibilityPresentationSettings	UI	Source/Engine/UI/Accessibility.hpp	233
AccessibilityProfile	UI	Source/Engine/UI/Accessibility.hpp	272
AccessibilitySettings	UI	Source/Engine/UI/Accessibility.hpp	217
AudioSystem	UI	Source/Engine/UI/Accessibility.hpp	26
AvSpeechTts	UI	Source/Engine/UI/Tts.hpp	60
BuildStatus	UI	Source/Engine/UI/PinkUI.hpp	1440
CardBoardComponent	UI	Source/Engine/UI/CardBoard.hpp	49
CardBoardSystem	UI	Source/Engine/UI/CardBoard.hpp	74
ColorMatrix3	UI	Source/Engine/UI/Accessibility.hpp	42
ColorVisionDeficiency	UI	Source/Engine/UI/Accessibility.hpp	34
Dropdown	UI	Source/Engine/UI/PinkUI.hpp	534
FlashGuard	UI	Source/Engine/UI/Accessibility.hpp	59
FocusDirection	UI	Source/Engine/UI/PinkUI.hpp	211
FocusPolicy	UI	Source/Engine/UI/PinkUI.hpp	187
FontAtlas	UI	Source/Engine/UI/	101

		FontAtlas.hpp	
FontAtlasHeader	UI	Source/Engine/UI/FontAtlas.hpp	45
FontAtlasSet	UI	Source/Engine/UI/FontAtlas.hpp	167
FontBuildOptions	UI	Source/Engine/UI/FontAtlas.hpp	25
FontBuildOptions	UI	Source/Engine/UI/FontBuilder.hpp	15
FontCoverage	UI	Source/Engine/UI/FontCoverage.hpp	21
FontFamilyFile	UI	Source/Engine/UI/FontAtlas.hpp	74
FontFamilyManifest	UI	Source/Engine/UI/FontFamilyAsset.hpp	17
FontPixelFormat	UI	Source/Engine/UI/FontAtlas.hpp	37
FontStyle	UI	Source/Engine/UI/FontAtlas.hpp	65
FrameOutput	UI	Source/Engine/UI/PinkUI.hpp	1418
GlyphRecord	UI	Source/Engine/UI/FontAtlas.hpp	80
GridView	UI	Source/Engine/UI/PinkUI.hpp	424
HyperlinkRange	UI	Source/Engine/UI/Hypertext.hpp	51
HypertextComponent	UI	Source/Engine/UI/Hypertext.hpp	32
HypertextFlags	UI	Source/Engine/UI/Hypertext.hpp	21
HypertextPool	UI	Source/Engine/UI/Hypertext.hpp	74
HypertextView	UI	Source/Engine/UI/Hypertext.hpp	67
ITextShaper	UI	Source/Engine/UI/PinkUI.hpp	33
LayoutCache	UI	Source/Engine/UI/PinkUI.hpp	736
LazyFontFamily	UI	Source/Engine/UI/FontFamilyAsset.hpp	101

ListView	UI	Source/Engine/UI/ PinkUI.hpp	560
LocalizationTable	UI	Source/Engine/UI/ Accessibility.hpp	24
LocalizationTable	UI	Source/Engine/UI/ Hypertext.hpp	16
LocalizationTable	UI	Source/Engine/UI/ PinkUI.hpp	39
NarrativeArchetype	UI	Source/Engine/UI/ NarrativeTree.hpp	23
NarrativeEvent	UI	Source/Engine/UI/ Accessibility.hpp	83
NarrativeGroup	UI	Source/Engine/UI/ NarrativeTree.hpp	37
NarrativeLocale	UI	Source/Engine/UI/ NarrativeTree.hpp	17
NarrativeLog	UI	Source/Engine/UI/ Accessibility.hpp	99
NarrativeNode	UI	Source/Engine/UI/ NarrativeTree.hpp	47
NarrativeTree	UI	Source/Engine/UI/ NarrativeTree.hpp	55
OledShiftState	UI	Source/Engine/UI/ PinkUI.hpp	463
PinkSdfDraw	UI	Source/Engine/UI/ PinkSdf.hpp	74
PinkSdfFlags	UI	Source/Engine/UI/ PinkSdf.hpp	24
PinkSdfGpuPrimitive	UI	Source/Engine/UI/ PinkSdf.hpp	51
PinkSdfPrimitive	UI	Source/Engine/UI/ PinkSdf.hpp	33
PinkSdfShape	UI	Source/Engine/UI/ PinkSdf.hpp	15
RadialProgress	UI	Source/Engine/UI/ PinkUI.hpp	584
RetainedFont	UI	Source/Engine/UI/ FontAtlas.hpp	293
SapiTts	UI	Source/Engine/UI/ Tts.hpp	30
ScrollView	UI	Source/Engine/UI/	491

		PinkUI.hpp	
SlotComponent	UI	Source/Engine/UI/CardBoard.hpp	27
SlotFlags	UI	Source/Engine/UI/CardBoard.hpp	15
SlotLayout	UI	Source/Engine/UI/CardBoard.hpp	65
SoundCueCategory	UI	Source/Engine/UI/SoundCues.hpp	26
SoundCueDef	UI	Source/Engine/UI/SoundCues.hpp	62
SoundCueRegistry	UI	Source/Engine/UI/SoundCues.hpp	71
SoundTrigger	UI	Source/Engine/UI/Accessibility.hpp	128
StateTrack	UI	Source/Engine/UI/Accessibility.hpp	417
SubtitleLine	UI	Source/Engine/UI/Accessibility.hpp	139
SubtitleQueue	UI	Source/Engine/UI/Accessibility.hpp	163
SystemContext	UI	Source/Engine/UI/PinkUI.hpp	28
TabBar	UI	Source/Engine/UI/PinkUI.hpp	515
TextAlign	UI	Source/Engine/UI/PinkUI.hpp	222
TextShapeParams	UI	Source/Engine/UI/PinkUI.hpp	34
TextVerticalAlign	UI	Source/Engine/UI/PinkUI.hpp	229
TimedSoundCue	UI	Source/Engine/UI/Accessibility.hpp	152
TooltipPlacer	UI	Source/Engine/UI/PinkUI.hpp	595
TtsBackend	UI	Source/Engine/UI/Accessibility.hpp	27
TtsBackend	UI	Source/Engine/UI/Tts.hpp	19
TtsPriority	UI	Source/Engine/UI/Tts.hpp	11

UiPalette	UI	Source/Engine/UI/UiTheme.hpp	41
UiPaletteSlot	UI	Source/Engine/UI/UiTheme.hpp	18
UiThemeAsset	UI	Source/Engine/UI/UiTheme.hpp	46
VirtualFileSystem	UI	Source/Engine/UI/FontAtlas.hpp	19
VoiceCaptionEvent	UI	Source/Engine/UI/Accessibility.hpp	193
VoiceCaptionProvider	UI	Source/Engine/UI/Accessibility.hpp	201
WidgetCanvas	UI	Source/Engine/UI/Accessibility.hpp	30
WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	290
WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	483
WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	607
WidgetCanvasSystem	UI	Source/Engine/UI/PinkUI.hpp	1453
WidgetComponent	UI	Source/Engine/UI/PinkUI.hpp	251
WidgetComponentFlags	UI	Source/Engine/UI/PinkUI.hpp	263
WidgetDesc	UI	Source/Engine/UI/PinkUI.hpp	309
WidgetFlags	UI	Source/Engine/UI/PinkUI.hpp	145
WidgetHandle	UI	Source/Engine/UI/PinkUI.hpp	80
WidgetHyperlink	UI	Source/Engine/UI/PinkUI.hpp	300
WidgetKind	UI	Source/Engine/UI/PinkUI.hpp	99
WidgetLayoutAsset	UI	Source/Engine/UI/WidgetLayoutAsset.hpp	15
WidgetPaintContext	UI	Source/Engine/UI/PinkUI.hpp	291
WidgetPaintContext	UI	Source/Engine/UI/	391

		PinkUI.hpp	

Engine/Video (20)

Symbol	Kind	Header	Line
AssetDatabase	Video	Source/Engine/Video/VideoStream.hpp	23
ClusterScan	Video	Source/Engine/Video/WebmDemuxer.hpp	108
ContainerKind	Video	Source/Engine/Video/WebmDemuxer.hpp	150
DecodeStatus	Video	Source/Engine/Video/VideoTypes.hpp	29
DemuxStatus	Video	Source/Engine/Video/WebmDemuxer.hpp	52
Element	Video	Source/Engine/Video/WebmDemuxer.hpp	85
FrameBuffer	Video	Source/Engine/Video/VideoPlayer.hpp	118
FrameSlot	Video	Source/Engine/Video/VideoPlayer.hpp	125
MappedFile	Video	Source/Engine/Video/VideoPlayer.hpp	22
RHI	Video	Source/Engine/Video/VideoPlayer.hpp	27
VideoPlayer	Video	Source/Engine/Video/VideoPlayer.hpp	33
VideoStreamComponent	Video	Source/Engine/Video/VideoStream.hpp	31
VideoStreamFlags	Video	Source/Engine/Video/VideoTypes.hpp	20
VideoSystem	Video	Source/Engine/Video/	49

		VideoStream.hpp	
VirtualFileSystem	Video	Source/Engine/ Video/ VideoStream.hpp	20
Vp9Backend	Video	Source/Engine/ Video/ Vp9Backend.hpp	30
WebmDemuxer	Video	Source/Engine/ Video/ WebmDemuxer.hpp	60
WebmFrame	Video	Source/Engine/ Video/ WebmDemuxer.hpp	38
WebmTrackInfo	Video	Source/Engine/ Video/ WebmDemuxer.hpp	14
YuvFrameView	Video	Source/Engine/ Video/ Vp9Backend.hpp	12

Engine/Weather (12)

Symbol	Kind	Header	Line
ClothSystem	Weather	Source/Engine/ Weather/Weather.hpp	13
WeatherAtmosphere Pass	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	128
WeatherContextCom ponent	Weather	Source/Engine/ Weather/Weather.hpp	32
WeatherDesc	Weather	Source/Engine/ Weather/Weather.hpp	91
WeatherFlags	Weather	Source/Engine/ Weather/Weather.hpp	19
WeatherGpuConstan ts	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	46
WeatherGpuResourc es	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	18
WeatherGpuStats	Weather	Source/Engine/	68

		Weather/ WeatherPreconciliated .hpp	
WeatherPreconciliatedDesc	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	32
WeatherPreconciliatedSystem	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	78
WeatherSystem	Weather	Source/Engine/ Weather/Weather.hpp	99
WeatherWindField	Weather	Source/Engine/ Weather/Weather.hpp	75

21. Appendix B - macros and platform defines

Macro	Header	Line
FBY_AOT_EXPORT	Source/Engine/Script/ AotAbi.hpp	396
FBY_AOT_EXPORT	Source/Engine/Script/ AotAbi.hpp	398
FBY_ASSERT	Source/Engine/Core/ Assert.hpp	13
FBY_ASSERT	Source/Engine/Core/ Assert.hpp	31
FBY_ASSERT_MSG	Source/Engine/Core/ Assert.hpp	22
FBY_ASSERT_MSG	Source/Engine/Core/ Assert.hpp	32
FBY_DEFINE_PLUGIN	Source/Engine/Core/ Plugin.hpp	56
FBY_DETAIL_CONCAT	Source/Engine/Core/ Definitions.hpp	261
FBY_DETAIL_CONCAT_INNER	Source/Engine/Core/ Definitions.hpp	260
FBY_FIELD	Source/Engine/Core/ Mirror.hpp	284
FBY_FLUID_HAS_MDSPAN	Source/Engine/Physics/ Fluid.hpp	16
FBY_FLUID_HAS_MDSPAN	Source/Engine/Physics/	18

	Fluid.hpp	
FBY_HAS_MDSPAN	Source/Engine/ECS/ Skeleton.hpp	17
FBY_HAS_MDSPAN	Source/Engine/ECS/ Skeleton.hpp	19
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/ Mirror.hpp	102
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/ Mirror.hpp	105
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/ Mirror.hpp	109
FBY_HAS_STD_FLAT_MAP	Source/Engine/Core/ FlatMap.hpp	19
FBY_HAS_STD_FLAT_MAP	Source/Engine/Core/ FlatMap.hpp	21
FBY_HAS_STD_FLAT_MAP	Source/Engine/Core/ FlatMap.hpp	24
FBY_PLATFORM_ABI_VERSION	Source/Plugins/ PlatformService/PlatformAPI.h	11
FBY_PLATFORM_CALL	Source/Plugins/ PlatformService/PlatformAPI.h	15
FBY_PLATFORM_CALL	Source/Plugins/ PlatformService/PlatformAPI.h	18
FBY_PLATFORM_EXPORT	Source/Plugins/ PlatformService/PlatformAPI.h	14
FBY_PLATFORM_EXPORT	Source/Plugins/ PlatformService/PlatformAPI.h	17
FBY_PLUGIN_ABI_VERSION	Source/Engine/Core/ PluginAPI.h	28
FBY_PLUGIN_API_H	Source/Engine/Core/ PluginAPI.h	18
FBY_PLUGIN_COMPONENT	Source/Engine/Core/ Plugin.hpp	53
FBY_PLUGIN_ENTRY	Source/Engine/Core/ PluginAPI.h	115
FBY_PLUGIN_EXPORT	Source/Engine/Core/ PluginAPI.h	31
FBY_PLUGIN_EXPORT	Source/Engine/Core/ PluginAPI.h	33
FBY_PLUGIN_EXPORT	Source/Engine/Core/ PluginAPI.h	35

FBY_PLUGIN_EXPORT	Source/Engine/Core/PluginApi.hpp	11
FBY_PLUGIN_EXPORT	Source/Engine/Core/PluginApi.hpp	13
FBY_PLUGIN_INFO_SYMBOL	Source/Engine/Core/PluginAPI.h	107
FBY_PLUGIN_LOAD_SYMBOL	Source/Engine/Core/PluginAPI.h	108
FBY_PLUGIN_UNLOAD_SYMBOL	Source/Engine/Core/PluginAPI.h	109
FBY_PROJECT_ASSETS	Source/Engine/Assets/ProjectScaffold.hpp	123
FBY_REFLECT_COMPONENT	Source/Engine/Core/Mirror.hpp	291
FBY_REFLECT_COMPONENT_EMPTY	Source/Engine/Core/Mirror.hpp	303
FBY_REGISTER_TYPE	Source/Engine/Core/Definitions.hpp	264
FBY_RELEVANCE_HAS_MDSPAN	Source/Engine/Network/RelevanceGrid.hpp	19
FBY_RELEVANCE_HAS_MDSPAN	Source/Engine/Network/RelevanceGrid.hpp	21
FBY_RESTRICT	Source/Engine/Core/Definitions.hpp	26
FBY_RESTRICT	Source/Engine/Core/Definitions.hpp	28
FBY_SDF_SHAPE_FIELD	Source/Engine/Physics/SdfPrimitives.hpp	172
FBY_TYPE_NAME	Source/Engine/Core/Definitions.hpp	280
FBY_UI_HAS_MDSPAN	Source/Engine/UI/PinkUI.hpp	21
FBY_UI_HAS_MDSPAN	Source/Engine/UI/PinkUI.hpp	23
FBY_UNREACHABLE	Source/Engine/Core/Assert.hpp	45
FBY_VERIFY	Source/Engine/Core/Assert.hpp	36
FBY_VK_DECLARE	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	126
FBY_VK_DEVICE_FUNCTIONS	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	36
FBY_VK_GLOBAL_FUNCTIO	Source/Engine/RHI/Vulkan/	15

NS	VulkanLoader.hpp	
FBY_VK_INSTANCE_FUNCTIONS	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	20
FORCEINLINE	Source/Engine/Core/Definitions.hpp	17
FORCEINLINE	Source/Engine/Core/Definitions.hpp	19
FORCEINLINE	Source/Engine/Core/Definitions.hpp	21